

# ALGEBRA 2

## Unit 10

| Name: | Topic: | Main Ideas/Questions     | Notes/Examples                                                                                                                              |
|-------|--------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|       |        | <b>SEQUENCE</b>          |                                                                                                                                             |
|       |        | <b>FINITE Sequence</b>   |                                                                                                                                             |
|       |        | <b>INFINITE Sequence</b> |                                                                                                                                             |
|       |        | <b>TERM Notation</b>     | <ul style="list-style-type: none"><li>The first term in a sequence</li><li>Each subsequent term is a fixed number in the sequence</li></ul> |

| Name:        | Topic:        | Main Ideas/Questions | Notes/Examples                                                                                                                                        |          |        |              |               |
|--------------|---------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|--------------|---------------|
|              |               | <b>SERIES</b>        | <table border="1"><thead><tr><th>Sequence</th><th>Series</th></tr></thead><tbody><tr><td>{1, 2, 3, 4}</td><td>{3, 6, 9, 12}</td></tr></tbody></table> | Sequence | Series | {1, 2, 3, 4} | {3, 6, 9, 12} |
| Sequence     | Series        |                      |                                                                                                                                                       |          |        |              |               |
| {1, 2, 3, 4} | {3, 6, 9, 12} |                      |                                                                                                                                                       |          |        |              |               |
|              |               | <b>PARTIAL SUMS</b>  | Find the partial sum for each given sequence.<br>1. {1, 2, 3, 4, 5, ...}; find $S_5$<br>2. {4, ...}                                                   |          |        |              |               |

| Name: | Topic: | Main Ideas/Questions       | Notes/Examples                                                                                                                                                                                                     |
|-------|--------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |        | <b>ARITHMETIC Sequence</b> |                                                                                                                                                                                                                    |
|       |        | <b>Common DIFFERENCE</b>   |                                                                                                                                                                                                                    |
|       |        | <b>EXAMPLES</b>            | Determine whether the sequence is arithmetic. If yes, identify the common difference and give the next three terms.<br>1. {5, 8, 11, 14, 17, ...}<br>2. {23, 16, 9, 2, -5, ...}<br>4. {-6, -8, -10, -12, -14, ...} |

| Name: | Topic: | Main Ideas/Questions    | Notes/Examples                                                                                                                                                                                                                                                                                                                                     |
|-------|--------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |        | <b>GEOMETRIC Series</b> | A geometric series is the sum of the terms of a geometric sequence. To find the sum, use the formula:<br>$S_n = \frac{a_1(1 - r^n)}{1 - r}$ where $n$ is the number of terms, $a_1$ is the first term, and $r$ is the common ratio.<br>Find the indicated sum for each geometric series.<br>1. {4 + 20 + 100 + 500 + ...}; $S_{14}$<br>2. {4, ...} |
|       |        | <b>EXAMPLES</b>         |                                                                                                                                                                                                                                                                                                                                                    |

| ARITHMETIC, GEOMETRIC... or Neither?                                                                                                                                                                     |                                                 |                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------|
| Directions: Determine the type of sequence. Shade the boxes using the color codes below. For arithmetic and geometric sequences, identify the common difference or common ratio, then write the formula. |                                                 |                                                                            |
| Arithmetic Sequence: Purple                                                                                                                                                                              | Geometric Sequence: Light Green                 | Neither: Uncolored                                                         |
| 1. {4, -20, 100, -500, ...}                                                                                                                                                                              | 2. {15, 5, $\frac{5}{3}$ , $\frac{5}{9}$ , ...} | 3. {1, -2, 3, -4, ...}                                                     |
| 4. {-18, -11, -4, 3, ...}                                                                                                                                                                                | 5. {1, 8, 27, 64, ...}                          | 6. { $\frac{9}{2}$ , $\frac{5}{2}$ , $\frac{11}{2}$ , $\frac{3}{2}$ , ...} |

## SEQUENCES & SERIES

NOTES • HOMEWORK • QUIZZES • TEST

Created by: ALL THINGS ALGEBRA®

# Thank you SO MUCH for purchasing this product!

I hope you found this resource useful in your classroom! Please consider leaving feedback in [my TpT store](#) or email me at [allthingsalgebra@gmail.com](mailto:allthingsalgebra@gmail.com) with any questions or comments.

You can also find me here:



## TERMS OF USE

© 2012-2024 Gina Wilson (All Things Algebra®)

### LICENSING TERMS:

By purchasing this product, the purchaser receives a limited **individual license** to reproduce the product for use within their classroom. This license is not intended for use by organizations or multiple users, including but not limited to school districts, schools, or multiple teachers within a grade level. This license is **non-transferable**, meaning it can not be transferred from one teacher to another.

If other teachers in your department would like to use this product, additional licenses can be purchased from my TpT store. If your school or district is interested in purchasing transferable licenses to accommodate staff changes, they may contact me at [allthingsalgebra@gmail.com](mailto:allthingsalgebra@gmail.com) for a quote.

### COPYRIGHT TERMS:

No part of this resource may be uploaded to the internet in any form, including classroom/personal websites or network drives, unless the site is password protected and can only be accessed by students.

Thank you for  
respecting my work!

Gina



## Unit 10 – Sequences & Series: Sample Unit Outline

|              | TOPIC                                                                                            | HOMEWORK       |
|--------------|--------------------------------------------------------------------------------------------------|----------------|
| <b>DAY 1</b> | Introduction to Sequences, Series, and Summation Notation                                        | HW #1          |
| <b>DAY 2</b> | Arithmetic Sequences & Series                                                                    | HW #2          |
| <b>DAY 3</b> | <b>Quiz 10-1</b>                                                                                 | None           |
| <b>DAY 4</b> | Geometric Sequences                                                                              | HW #3          |
| <b>DAY 5</b> | Geometric Series & Infinite Geometric Series;<br>Finding the Sum of a Convergent Infinite Series | HW #4          |
| <b>DAY 6</b> | Arithmetic vs. Geometric (Mixed Review with Applications)                                        | HW #5          |
| <b>DAY 7</b> | Unit 10 Review                                                                                   | Study for Test |
| <b>DAY 8</b> | <b>UNIT 10 TEST</b>                                                                              | None           |

|        |        |
|--------|--------|
| Name:  | Date:  |
| Topic: | Class: |

| Main Ideas/Questions             | Notes/Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----|----|----|---|---|---|-------|---|---|---|----|----|
| SEQUENCE                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
| FINITE<br><i>Sequence</i>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
|                                  | Example:                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
| INFINITE<br><i>Sequence</i>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
|                                  | Example:                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
| TERM<br><i>Notation</i>          | <ul style="list-style-type: none"><li>The first term in a sequence is denoted _____.</li><li>Each subsequent term is denoted _____, where _____ is the term number in the sequence.</li></ul> <p>Example: Given {1, 5, 9, 13, 17, ...}, identify the following term values:</p> <p><math>a_1</math> : _____    <math>a_4</math> : _____    <math>a_9</math> : _____    <math>a_{12}</math> : _____</p>                                                                        |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
| SEQUENCES<br><i>as Functions</i> | <p>Since each term value is paired with exactly one term number, a sequence is a function with the following properties:</p> <p>The <b>domain</b> is the set of _____.</p> <table><tr><td><math>n</math></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td><math>a_n</math></td><td>1</td><td>5</td><td>9</td><td>13</td><td>17</td></tr></table> <p>The <b>range</b> is the set of _____.</p> <p>In an infinite set, the domain is the the set of _____</p> |                                                       | $n$ | 1  | 2  | 3 | 4 | 5 | $a_n$ | 1 | 5 | 9 | 13 | 17 |
|                                  | $n$                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1                                                     | 2   | 3  | 4  | 5 |   |   |       |   |   |   |    |    |
| $a_n$                            | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5                                                     | 9   | 13 | 17 |   |   |   |       |   |   |   |    |    |
| RECURSIVE<br><i>Formulas</i>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
|                                  | <p>The <b>Fibonacci Sequence</b>: _____</p> <p>In this sequence, <math>a_1 = 0</math>, <math>a_2 = 1</math>, then for each subsequent term,</p> <p>_____</p>                                                                                                                                                                                                                                                                                                                  |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
| EXAMPLES                         | Find the first 5 terms of each sequence.                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                       |     |    |    |   |   |   |       |   |   |   |    |    |
|                                  | 1. $a_1 = 14$ ; $a_n = a_{n-1} + 9$ (for $n \geq 2$ )                                                                                                                                                                                                                                                                                                                                                                                                                         | 2. $a_1 = 6$ ; $a_n = 2a_{n-1} - 5$ (for $n \geq 2$ ) |     |    |    |   |   |   |       |   |   |   |    |    |

|                                    |                                                                         |                                                                                |
|------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|                                    | <b>3.</b> $a_1 = 354$ ; $a_n = \frac{a_{n-1} + 6}{3}$ (for $n \geq 2$ ) | <b>4.</b> $a_1 = 3, a_2 = 4$ ; $a_n = a_{n-1} \cdot a_{n-2}$ (for $n \geq 3$ ) |
|                                    | <b>Write a rule for each sequence. Then give the next 3 terms.</b>      |                                                                                |
|                                    | <b>5.</b> $\{4, 11, 32, 95, 284, \dots\}$                               | <b>6.</b> $\{100, 60, 40, 30, 25, \dots\}$                                     |
| <b>EXPLICIT</b><br><i>Formulas</i> |                                                                         |                                                                                |
|                                    |                                                                         |                                                                                |
| <b>EXAMPLES</b>                    | <b>Find the first 5 terms of each sequence.</b>                         |                                                                                |
|                                    | <b>7.</b> $a_n = 7(n - 3)$                                              | <b>8.</b> $a_n = \left(\frac{1}{3}\right)^{n+1}$                               |
|                                    | <b>9.</b> $a_n = -4n^2$                                                 | <b>10.</b> $a_n = 2n(n + 5)$                                                   |
|                                    | <b>11.</b> $a_n = (-2)^{n-1}$                                           | <b>12.</b> $a_n = (n - 4)^2$                                                   |
|                                    | <b>Write a rule for each sequence. Then give the next 3 terms.</b>      |                                                                                |
|                                    | <b>13.</b> $\{-7, -9, -11, -13, -15, \dots\}$                           | <b>14.</b> $\{3, 6, 11, 18, 27, \dots\}$                                       |
|                                    | <b>15.</b> $\left\{2, \frac{5}{2}, 3, \frac{7}{2}, 4, \dots\right\}$    | <b>16.</b> $\{1, \sqrt{2}, \sqrt{3}, 2, \sqrt{5}, \dots\}$                     |

|                              |                                                                                                                                  |                         |                                                                                               |                                                                             |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Name:                        |                                                                                                                                  | Date:                   |                                                                                               |                                                                             |
| Topic:                       |                                                                                                                                  | Class:                  |                                                                                               |                                                                             |
| Main Ideas/Questions         | Notes/Examples                                                                                                                   |                         |                                                                                               |                                                                             |
| SERIES                       |                                                                                                                                  |                         |                                                                                               |                                                                             |
|                              | Sequence                                                                                                                         | {1, 2, 3, 4}            | {3, 6, 9, 12,...}                                                                             | $\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots\right\}$ |
|                              | Series                                                                                                                           |                         |                                                                                               |                                                                             |
| PARTIAL SUMS<br><div></div>  | Find the partial sum for each given sequence.                                                                                    |                         |                                                                                               |                                                                             |
|                              | 1. {1, 2, 3, 4, 5,...}; find $s_9$                                                                                               |                         | 2. {4, 7, 13, 25, 49,...}; find $s_4$                                                         |                                                                             |
|                              | 3. {1, 4, 9, 16, 25,...}; find $s_6$                                                                                             |                         | 4. $\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots\right\}$ ; find $s_3$ |                                                                             |
| SUMMATION<br><i>Notation</i> | A way to represent a series using the greek letter $\Sigma$ to denote the sum.                                                   |                         |                                                                                               |                                                                             |
|                              | <div><div></div><div></div><div><math>\sum_{n=1}^5 2n</math></div><div></div></div> <div>Find the sum of the series above:</div> |                         |                                                                                               |                                                                             |
| EXAMPLES                     | Expand each series and evaluate.                                                                                                 |                         |                                                                                               |                                                                             |
|                              | 5. $\sum_{n=1}^{12} (n - 1)$                                                                                                     |                         |                                                                                               |                                                                             |
|                              |                                                                                                                                  | 6. $\sum_{n=1}^7 (-3n)$ |                                                                                               |                                                                             |

$$7. \sum_{k=1}^4 (5k + 2)$$

$$8. \sum_{a=2}^9 (2a - 7)$$

$$9. \sum_{y=1}^5 (y^2 + y)$$

$$10. \sum_{c=2}^8 \left( \frac{c^3}{2} \right)$$

$$11. \sum_{m=2}^6 (m^2 - 1)$$

$$12. 3 \cdot \sum_{p=4}^8 (4 - p)^2$$

$$13. \frac{1}{2} \cdot \sum_{x=2}^4 (x^2 - 5)$$



Name: \_\_\_\_\_

Unit 10: Sequences & Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 1: Intro to Sequences & Series

**\*\* This is a 2-page document! \*\***

**Find the first five terms of each sequence.**

1.  $a_1 = 9$ ;  $a_n = a_{n-1} - 2$  (for  $n \geq 2$ )

2.  $a_1 = -4$ ;  $a_n = -3a_{n-1} + 1$  (for  $n \geq 2$ )

3.  $a_1 = 120$ ;  $a_n = \frac{1}{2}a_{n-1} + 8$  (for  $n \geq 2$ )

4.  $a_1 = 2$ ,  $a_2 = 5$ ;  $a_n = a_{n-1} \cdot a_{n-2} - 4$  (for  $n \geq 3$ )

**Write a possible recursive rule, then find  $a_7$ .**

5.  $\{7, 12, 17, 22, 27, \dots\}$

6.  $\{2, 3, 8, 19, 46, \dots\}$

**Find the first five terms of each sequence.**

7.  $a_n = -2(n - 7)$

8.  $a_n = \left(\frac{1}{2}\right)^{n+3}$

9.  $a_n = 4n^2$

10.  $a_n = (n + 1)^3$

**Write a possible explicit rule, then find  $a_{10}$ .**

11.  $\{-7, -6, -5, -4, -3, \dots\}$

12.  $\{5, 7, 9, 11, 13, \dots\}$

13.  $\{0, 7, 26, 63, 124, \dots\}$

14.  $\left\{\frac{1}{3}, \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{5}{7}, \dots\right\}$

**Find the partial sum for each sequence.**

15.  $\{38, 31, 24, 17, \dots\}; S_{12}$

16.  $\{1, 8, 27, 64, 125, \dots\}; S_7$



**Expand each series and evaluate.**

17.  $\sum_{n=1}^{14} (n - 6)$

18.  $\sum_{p=1}^9 (4p)$

19.  $\sum_{m=1}^5 (-2m + 7)$

20.  $\sum_{y=1}^8 (3y^2 - 2y)$

21.  $\sum_{c=3}^{10} \left( \frac{c^2 + 1}{2} \right)$

22.  $\sum_{n=2}^7 \left( \frac{n-2}{n+2} \right)$

23.  $2 \cdot \sum_{a=5}^{13} (a + 2)^2$

24.  $\frac{1}{4} \cdot \sum_{r=3}^{11} (15 - 2r)$

|        |        |
|--------|--------|
| Name:  | Date:  |
| Topic: | Class: |

| Main Ideas/Questions                                   | Notes/Examples                                                                                                                                      |                                            |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>ARITHMETIC</b><br><i>Sequence</i>                   |                                                                                                                                                     |                                            |
|                                                        |                                                                                                                                                     |                                            |
| <i>Common</i><br><b>DIFFERENCE</b>                     |                                                                                                                                                     |                                            |
|                                                        |                                                                                                                                                     |                                            |
| <b>EXAMPLES</b>                                        | <b>Determine whether the sequence is arithmetic. If yes, identify the common difference and give the next three terms.</b>                          |                                            |
|                                                        | 1. { 5, 8, 11, 14, 17, ... }                                                                                                                        | 2. { 23, 16, 9, 2, - 5, ... }              |
|                                                        | 3. { 1, 2, 3, 5, 8, ... }                                                                                                                           | 4. { -6, - 8, - 10, - 12, - 14, ... }      |
|                                                        | 5. { -20, - 10, - 5, 0, 5, ... }                                                                                                                    | 6. { -35, - 23, - 11, 1, 13, ... }         |
| <b>ARITHMETIC</b><br><b>SEQUENCE</b><br><i>Formula</i> | The $n^{\text{th}}$ term of any arithmetic sequence can be found using the formula:<br><div></div> where $a_1$ is the _____ and<br>$d$ is the _____ |                                            |
| <b>EXAMPLES</b>                                        | <b>Write a rule for each sequence, then find the indicated term.</b>                                                                                |                                            |
|                                                        | 7. { -5, - 1, 3, 7, ... } ; $a_{24}$                                                                                                                | 8. { 20, 11, 2, - 7, - 16 ... } ; $a_{15}$ |
|                                                        | 9. { -6, - 9, - 12, - 15, ... } ; $a_{38}$                                                                                                          | 10. { 8, 9.5, 11, 12.5, ... } ; $a_{48}$   |

|                                    |                                                                                                                                                                                                                                                                                                                              |                                                     |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|                                    | If the given terms are part of an arithmetic sequence, write a rule for the sequence, then find $a_{12}$ .                                                                                                                                                                                                                   |                                                     |
|                                    | 11. $a_1 = 3$ and $a_3 = -7$                                                                                                                                                                                                                                                                                                 | 12. $a_1 = -17$ and $a_{24} = 6$                    |
| <b>ARITHMETIC</b><br><i>Series</i> | Recall that a series is the _____ of a sequence. When finding the sum of an <b>arithmetic series</b> , use the following formula:<br><div style="border: 1px solid black; border-radius: 15px; height: 60px; margin: 10px auto; width: 250px;"></div> where $n$ is the _____,<br>$a_1$ is the _____, and $a_n$ is the _____. |                                                     |
| <b>EXAMPLES</b>                    | Find the indicated sum for each arithmetic series.                                                                                                                                                                                                                                                                           |                                                     |
|                                    | 13. $\{2 + 4 + 6 + 8 + \dots\}; S_{10}$                                                                                                                                                                                                                                                                                      | 14. $\{(-6) + (-19) + (-32) + (-45) + \dots\}; S_8$ |
|                                    | 15. $\sum_{x=1}^{12} (6x - 4)$                                                                                                                                                                                                                                                                                               | 16. $\sum_{m=8}^{17} (-2m + 3)$                     |



Name: \_\_\_\_\_

Unit 10: Sequences & Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 2: Arithmetic Sequences & Series

**\*\* This is a 2-page document! \*\***

**Determine whether the sequence is arithmetic. If yes, identify the common difference and give the next three terms.**

|                                    |                                                                    |
|------------------------------------|--------------------------------------------------------------------|
| 1. $\{-3, -12, -48, -192, \dots\}$ | 2. $\{-18, -27, -36, -45, \dots\}$                                 |
| 3. $\{4, 11, 18, 25, \dots\}$      | 4. $\{16, 8, 4, 1, \dots\}$                                        |
| 5. $\{-9, -33, -57, -81, \dots\}$  | 6. $\{\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots\}$ |

**Write a rule for each sequence, then find the indicated term.**

|                                            |                                         |
|--------------------------------------------|-----------------------------------------|
| 7. $\{12, 27, 42, 57, \dots\}; a_{18}$     | 8. $\{29, 9, -11, -31, \dots\}; a_{32}$ |
| 9. $\{-40, -44, -48, -52, \dots\}; a_{49}$ | 10. $\{50, 39, 28, 17, \dots\}; a_{54}$ |

11. Karen's New Year's resolution is to work out each day on her elliptical. The number of calories she has burned each day since she started is shown below. Write a rule for the sequence, and find the number of calories that Karen will burn on the 30<sup>th</sup> day.

$\{349, 357, 365, 373, 381, \dots\}$

**If the given terms are part of an arithmetic sequence, write a rule for the sequence, then find  $a_9$ .**

**12.**  $a_1 = 11$  and  $a_5 = 27$

**13.**  $a_1 = 3$  and  $a_{14} = -23$

**14.**  $a_1 = -5$  and  $a_7 = 91$

**15.**  $a_1 = \frac{1}{2}$  and  $a_{23} = 6$

**Find the indicated sum for each arithmetic series.**

**16.**  $\{4 + 6 + 8 + 10 + \dots\}; S_{21}$

**17.**  $\{24 + 31 + 38 + 45 + \dots\}; S_9$

**18.**  $\{15 + 4 + (-7) + (-18) + \dots\}; S_{12}$

**19.**  $\sum_{k=1}^{25} (5k + 1)$

**20.**  $\sum_{n=9}^{14} (17 - 6n)$

**21.**  $\sum_{x=11}^{35} (3x - 12)$

Name: \_\_\_\_\_

**Algebra 2**

Date: \_\_\_\_\_ Per: \_\_\_\_\_

**Unit 10: Sequences & Series**

**Quiz 10-1:** Intro to Seq. & Ser., Arithmetic Seq. & Ser.

**Part I:** Intro to Sequences & Series

**Give the first five terms of each sequence.**

1.  $a_1 = -4; a_n = a_{n-1} - 6$  (for  $n \geq 2$ )

1. \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

2.  $a_1 = 3, a_2 = 8; a_n = 2a_{n-2} + a_{n-1}$  (for  $n \geq 3$ )

2. \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

3.  $a_n = 15 - 2n$

3. \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

4.  $a_n = n^3 - 20$

4. \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

5. Write a recursive formula to represent the sequence, then find  $a_7$ .

5. \_\_\_\_\_

$\{4, 11, 32, 95, 284, \dots\}$

$a_7 =$  \_\_\_\_\_

6. Write an explicit formula to represent the sequence, then find  $a_{12}$ .

6. \_\_\_\_\_

$\{8, 11, 16, 23, 32, \dots\}$

$a_{12} =$  \_\_\_\_\_

**Find the partial sum for each sequence.**

7.  $\{2, 3, 5, 8, 13, \dots\}; S_8$

7. \_\_\_\_\_

8.  $\{0, -1, -3, -6, -10, \dots\}; S_{14}$

8. \_\_\_\_\_

**Expand and evaluate the series.**

9.  $\sum_{a=1}^{10} (a-2)^2$

9. \_\_\_\_\_

**Part II: Arithmetic Sequences & Series**

**Determine whether the sequence is arithmetic. If yes, identify the common difference.**

10.  $\{1, 2, 3, 5, 8, \dots\}$  10. \_\_\_\_\_

11.  $\{17, 9, 1, -7, -15, \dots\}$  11. \_\_\_\_\_

12.  $\left\{\frac{4}{5}, \frac{6}{5}, \frac{8}{5}, 2, \frac{12}{5}, \dots\right\}$  12. \_\_\_\_\_

**Write a rule for each sequence, then find the indicated term.**

13.  $\{-7, -4, -1, 2, \dots\}; a_{38}$  13. \_\_\_\_\_

$a_{38} =$  \_\_\_\_\_

14.  $\{12, 10.5, 9, 7.5, \dots\}; a_{55}$  14. \_\_\_\_\_

$a_{57} =$  \_\_\_\_\_

15. The terms below are part of an arithmetic sequence.  
Write the rule, then find  $a_9$ . 15. \_\_\_\_\_

$a_1 = -27$  and  $a_{18} = 41$   $a_9 =$  \_\_\_\_\_

**Find the indicated sum for each arithmetic series.**

16.  $\{(-2) + (-7) + (-12) + (-17) + \dots\}; S_{27}$  16. \_\_\_\_\_

17.  $\sum_{m=1}^{31} (7m - 3)$  17. \_\_\_\_\_

18.  $\sum_{k=5}^{23} (1 - 2k)$  18. \_\_\_\_\_

|                                                       |                                                                                                               |        |                                 |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|---------------------------------|
| Name:                                                 |                                                                                                               | Date:  |                                 |
| Topic:                                                |                                                                                                               | Class: |                                 |
| Main Ideas/Questions                                  | Notes/Examples                                                                                                |        |                                 |
| <b>GEOMETRIC</b><br><i>Sequence</i>                   |                                                                                                               |        |                                 |
|                                                       |                                                                                                               |        |                                 |
| <i>Common</i><br><b>RATIO</b>                         |                                                                                                               |        |                                 |
|                                                       |                                                                                                               |        |                                 |
| <b>EXAMPLES</b>                                       | Determine whether the sequence is geometric. If yes, identify the common ratio and give the next three terms. |        |                                 |
|                                                       | 1. {6, 12, 24, 48, ...}                                                                                       |        | 2. {810, 270, 90, 30, ...}      |
|                                                       | 3. {-2, 10, -50, 250, ...}                                                                                    |        | 4. {4, 8, 20, 60, ...}          |
|                                                       | 5. $\left\{4, -1, \frac{1}{4}, -\frac{1}{16}, \dots\right\}$                                                  |        | 6. {2, 4, 16, 256, ...}         |
|                                                       | 7. {-2, -14, -98, -686, ...}                                                                                  |        | 8. {3.2, 8, 20, 50, ...}        |
| <b>GEOMETRIC</b><br><b>SEQUENCE</b><br><i>Formula</i> | The $n^{\text{th}}$ term of any geometric sequence can be found using the formula:                            |        |                                 |
|                                                       | <div></div> <p>where <math>a_1</math> is the _____ and<br/><math>r</math> is the _____</p>                    |        |                                 |
| <b>EXAMPLES</b>                                       | Write a rule for each sequence, then find $a_6$ .                                                             |        |                                 |
|                                                       | 9. {1, 5, 25, 125, ...}                                                                                       |        | 10. {130, 65, 32.5, 16.25, ...} |



|                    |                                                                                                                                                              |                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
|                    | 11. $\{2, -8, 32, -128, \dots\}$                                                                                                                             | 12. $\{8, 24, 72, 216, \dots\}$              |
|                    | 13. $\{135, 90, 60, 40, \dots\}$                                                                                                                             | 14. $\{-3200, 800, -200, 50, \dots\}$        |
|                    | <b>Use the information given to find the indicated value.</b>                                                                                                |                                              |
|                    | 15. $a_1 = 7$ and $r = 4$ ; find $a_8$                                                                                                                       | 16. $a_5 = -68$ and $r = -0.5$ ; find $a_1$  |
|                    | 17. $a_1 = 13$ and $a_4 = 4459$ ; find $a_3$                                                                                                                 | 18. $a_2 = 6$ and $a_5 = 0.048$ ; find $a_4$ |
|                    |                                                                                                                                                              |                                              |
| <b>APPLICATION</b> | Last year, Jackson bought a brand new car for \$47,500. If the car depreciates in value by 20% each year, what will the car be worth when it is 8 years old? |                                              |



Name: \_\_\_\_\_

Unit 10: Sequences & Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 3: Geometric Sequences

**\*\* This is a 2-page document! \*\***

**Determine whether the sequence is geometric. If yes, identify the common ratio and give the next three terms.**

1.  $\{2, 16, 128, \dots\}$

2.  $\{2700, -900, 300, \dots\}$

3.  $\{5, 10, 30, \dots\}$

4.  $\{-3, 12, -48, \dots\}$

5.  $\left\{-1, -\frac{1}{2}, -\frac{1}{4}, \dots\right\}$

6.  $\{8, 9, 10, \dots\}$

**Write a rule for each sequence, then find  $a_8$ .**

7.  $\{9, -18, 36, -72, \dots\}$

8.  $\{3, 15, 75, 375, \dots\}$

9.  $\{6400, 1600, 400, 100, \dots\}$

10.  $\{-1.5, 4.5, -13.5, 40.5, \dots\}$

11.  $\{-2, -14, -98, -686, \dots\}$

12.  $\left\{\frac{16}{81}, \frac{8}{27}, \frac{4}{9}, \frac{2}{3}, \dots\right\}$

Use the information given to write a rule for the sequence and find the indicated value.

13.  $a_1 = 4$  and  $r = 6$ ; find  $a_7$

14.  $a_8 = 1024$  and  $r = -2$ ; find  $a_4$

15.  $a_3 = -27$  and  $r = \frac{1}{3}$ ; find  $a_1$

16.  $a_1 = 2400$  and  $a_4 = 37.5$ ; find  $a_2$

17.  $a_2 = 48$  and  $a_4 = 75$ ; find  $a_1$

18.  $a_3 = 16$  and  $a_6 = 250$ ; find  $a_8$

19. Jameson opened a new savings account with \$1500. The balance of the account for the first three months is given in the table below. Write a rule to represent the balance of her account, then estimate the balance of the account in 2 years.

| Month | Balance (\$) |
|-------|--------------|
| 1     | 1500         |
| 2     | 1545         |
| 3     | 1591.35      |

20. Brady dropped a bouncy ball from a height of 64 feet. After the second bounce, the ball reached a height of 36 feet. Write a rule to represent the height of the ball after each bounce, then find the height after the fourth bounce.

|        |        |
|--------|--------|
| Name:  | Date:  |
| Topic: | Class: |

| Main Ideas/Questions       | Notes/Examples                                                                                                                                                                                                                                      |                                                              |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| GEOMETRIC<br><i>Series</i> | A <b>geometric series</b> is the _____ of a geometric sequence.<br>To find the sum, use the following formula:<br><div></div><br>where <b><i>n</i></b> is the _____,<br><b><i>a</i><sub>1</sub></b> is the _____, and <b><i>r</i></b> is the _____. |                                                              |
|                            | Find the indicated sum for each geometric series.                                                                                                                                                                                                   |                                                              |
| EXAMPLES                   | 1. {4 + 20 + 100 + 500 + ...}; <i>S</i> <sub>14</sub>                                                                                                                                                                                               | 2. {81 + 27 + 9 + 3 + ...}; <i>S</i> <sub>20</sub>           |
|                            | 3. {3 + 6 + 12 + 24 + ... + 6144}                                                                                                                                                                                                                   | 4. {4 – 12 + 36 – 108 + ... + 2916}                          |
|                            | 5. $\sum_{n=1}^{25} 2^{n-1}$                                                                                                                                                                                                                        | 6. $\sum_{x=1}^8 -32 \cdot \left(-\frac{1}{2}\right)^{x-1}$  |
|                            | 7. $\sum_{k=6}^{20} -4 \cdot (-3)^{k-1}$                                                                                                                                                                                                            | 8. $\sum_{m=2}^{11} 81 \cdot \left(\frac{1}{2}\right)^{m-1}$ |

|                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--|-------|--|-------|--|-------|--|-------|--|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--|-------|--|-------|--|-------|--|-------|--|-------|--|
| <h1>INFINITE</h1> <p>Geometric Series</p>                                                                                          | Find the partial sums for each infinite series below:                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
|                                                                                                                                    | $\left\{ \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \right\}$ <table border="1"> <tr><td><math>S_1</math></td><td></td></tr> <tr><td><math>S_2</math></td><td></td></tr> <tr><td><math>S_3</math></td><td></td></tr> <tr><td><math>S_4</math></td><td></td></tr> <tr><td><math>S_5</math></td><td></td></tr> <tr><td><math>S_6</math></td><td></td></tr> </table> <p>A series that approaches a certain sum is called a <b>CONVERGENT SERIES</b>.</p> | $S_1$                                                                                                     |  | $S_2$ |  | $S_3$ |  | $S_4$ |  | $S_5$ |  | $S_6$ |  | $\left\{ \frac{1}{2} + 1 + 2 + 4 + 8 + \dots \right\}$ <table border="1"> <tr><td><math>S_1</math></td><td></td></tr> <tr><td><math>S_2</math></td><td></td></tr> <tr><td><math>S_3</math></td><td></td></tr> <tr><td><math>S_4</math></td><td></td></tr> <tr><td><math>S_5</math></td><td></td></tr> <tr><td><math>S_6</math></td><td></td></tr> </table> <p>A series that does not have a certain sum is called a <b>DIVERGENT SERIES</b>.</p> | $S_1$ |  | $S_2$ |  | $S_3$ |  | $S_4$ |  | $S_5$ |  | $S_6$ |  |
|                                                                                                                                    | $S_1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
|                                                                                                                                    | $S_2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_3$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_4$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_5$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_6$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_1$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_2$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_3$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_4$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_5$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| $S_6$                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| <ul style="list-style-type: none"> <li>If _____, then the series is _____.</li> <li>If _____, then the series is _____.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| <p>Convergent Series</p> <h2>FORMULA</h2>                                                                                          | <p>To find the sum of a <b>convergent infinite geometric series</b>, use the formula:</p> <div style="border: 1px solid black; height: 50px; width: 100%;"></div>                                                                                                                                                                                                                                                                                                         |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
| <h2>EXAMPLES</h2>                                                                                                                  | Determine if the series is convergent or divergent. If convergent, find the sum.                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
|                                                                                                                                    | <p>9. <math>\left\{ 1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \right\}</math></p>                                                                                                                                                                                                                                                                                                                                                                             | <p>10. <math>\left\{ \frac{2}{3} + \frac{8}{3} + \frac{32}{3} + \frac{128}{3} + \dots \right\}</math></p> |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
|                                                                                                                                    | <p>11. <math>\{ 5.8 + 3.48 + 2.088 + 1.2528 + \dots \}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                         | <p>12. <math>\{ 128 - 96 + 72 - 54 + \dots \}</math></p>                                                  |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
|                                                                                                                                    | <p>13. <math>\sum_{x=1}^{\infty} -2 \cdot \left( \frac{1}{4} \right)^{x-1}</math></p>                                                                                                                                                                                                                                                                                                                                                                                     | <p>14. <math>\sum_{p=1}^{\infty} 7 \cdot (0.8)^{p-1}</math></p>                                           |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |
|                                                                                                                                    | <p>15. <math>\sum_{w=4}^{\infty} 8 \cdot \left( -\frac{5}{2} \right)^{w-1}</math></p>                                                                                                                                                                                                                                                                                                                                                                                     | <p>16. <math>\sum_{k=2}^{\infty} -6 \cdot \left( -\frac{3}{4} \right)^{k-1}</math></p>                    |  |       |  |       |  |       |  |       |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |  |       |  |       |  |       |  |       |  |       |  |

Name: \_\_\_\_\_

Unit 10: Sequences &amp; Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 4: Geometric Series

**\*\* This is a 2-page document! \*\***

Find the indicated sum for each geometric series.

1.  $\{2 + 6 + 18 + 54 + \dots\}; S_{12}$

2.  $\left\{18 - 9 + \frac{9}{2} - \frac{9}{4} + \dots\right\}; S_8$

3.  $\left\{10 + 1 + \frac{1}{10} + \frac{1}{100} + \dots\right\}; S_7$

4.  $\left\{-\frac{1}{8} + \frac{1}{2} - 2 + 8 + \dots\right\}; S_{10}$

5.  $\{(-1) + (-6) + (-36) + \dots + (-7776)\}$

6.  $\{1875 + 750 + 300 + \dots + 3.072\}$

7.  $\sum_{x=1}^{12} (-4)^{x-1}$

8.  $\sum_{m=1}^7 24 \cdot \left(\frac{1}{2}\right)^{m-1}$

9.  $\sum_{k=4}^{14} 81 \cdot \left(-\frac{4}{3}\right)^{k-1}$

10.  $\sum_{w=2}^9 \frac{1}{10} \cdot (5)^{w-1}$

**Determine whether each geometric sequence is convergent or divergent.**

**Find the sum when possible.**

11.  $\left\{180 - 60 + 20 - \frac{20}{3} + \dots\right\}$

12.  $\{250 + 200 + 160 + 128 + \dots\}$

13.  $\left\{3 + \frac{9}{2} + \frac{27}{4} + \frac{81}{8} + \dots\right\}$

14.  $\left\{-\frac{8}{3} + \frac{4}{3} - \frac{2}{3} + \frac{1}{3} + \dots\right\}$

15.  $\sum_{k=1}^{\infty} -\frac{1}{4} \cdot \left(-\frac{1}{5}\right)^{k-1}$

16.  $\sum_{n=1}^{\infty} \frac{1}{2} \cdot 5^{n-1}$

17.  $\sum_{y=1}^{\infty} -6.8 \cdot \left(\frac{3}{5}\right)^{y-1}$

18.  $\sum_{m=1}^{\infty} 18 \cdot (0.25)^{m-1}$

19.  $\sum_{p=1}^{\infty} 10 \cdot \left(-\frac{4}{3}\right)^{p-1}$

20.  $\sum_{x=1}^{\infty} -128 \cdot \left(-\frac{1}{4}\right)^{x-1}$

# ARITHMETIC, GEOMETRIC... or Neither?

**Directions:** Determine the type of sequence. Shade the boxes using the color codes below.  
For arithmetic and geometric sequences, identify the common difference or common ratio, then write the formula.

**Arithmetic Sequence:** Purple

**Geometric Sequence:** Light Green

**Neither:** Uncolored

|                                      |                                                                  |                                                                      |
|--------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------|
| 1. $\{4, -20, 100, -500, \dots\}$    | 2. $\left\{15, 5, \frac{5}{3}, \frac{5}{9}, \dots\right\}$       | 3. $\{1, -2, 3, -4, \dots\}$                                         |
| 4. $\{-18, -11, -4, 3, \dots\}$      | 5. $\{1, 8, 27, 64, \dots\}$                                     | 6. $\left\{\frac{9}{4}, \frac{5}{2}, \frac{11}{4}, 3, \dots\right\}$ |
| 7. $\{-3, -27, -243, -2187, \dots\}$ | 8. $\left\{-5, -\frac{17}{2}, -12, -\frac{31}{2}, \dots\right\}$ | 9. $\left\{100, 40, 16, \frac{32}{5}, \dots\right\}$                 |

## SERIES REVIEW

| <b>Directions:</b> Indicate whether the series is arithmetic or geometric, then find the sum. |                                                                  |
|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 10. $\{3, -1, -5, -9, \dots\}; S_{26}$                                                        | 11. $\left\{-60, -30, -15, -\frac{15}{2}, \dots\right\}; S_{14}$ |
| 12. $\sum_{k=1}^9 -3 \cdot 2^{k-1}$                                                           | 13. $\sum_{x=4}^{39} (5x + 2)$                                   |



|                                                                    |                                                                |
|--------------------------------------------------------------------|----------------------------------------------------------------|
| 14. $\sum_{n=4}^{22} \left(1 - \frac{1}{3}n\right)$                | 15. $\sum_{m=2}^{10} -16 \cdot \left(\frac{1}{2}\right)^{m-1}$ |
| 16. $\sum_{y=1}^{\infty} -5 \cdot \left(-\frac{3}{4}\right)^{y-1}$ | 17. $\sum_{n=4}^{\infty} \frac{1}{3} \cdot 6^{n-1}$            |

## APPLICATIONS!

18. Ian is playing a video game. The number of points he has earned after every hour is shown below. Determine whether this represents an arithmetic or geometric sequence. Then find the approximate number of points earned after 9 hours. Round your answer to the nearest hundredth.

| Hour | Points |
|------|--------|
| 1    | 384    |
| 2    | 480    |
| 3    | 600    |
| 4    | 750    |

19. Mikayla is building a pyramid of paper cups. The bottom row has 61 cups while the top row has only one cup. If there are 16 total rows and the number of cups on each row represents an arithmetic sequence, find the total number of cups.

20. The amount of interest that Ben earned on his savings account for the past three years since opening the account is shown in the table below. Determine whether this pattern represents an arithmetic or geometric sequence. Then, find the total amount of interest that Ben will earn after 20 years.

| Year | Interest |
|------|----------|
| 1    | \$72     |
| 2    | \$79.20  |
| 3    | \$87.12  |

Name: \_\_\_\_\_

Unit 10: Sequences &amp; Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 5: Arithmetic & Geometric  
Sequences/Series (Mixed Review)**\*\* This is a 2-page document! \*\*****Determine whether the sequence is arithmetic, geometric, or neither. For arithmetic and geometric sequences, identify the common difference or common ratio, then write the formula.**

1.  $\{1, 2, 6, 24, 120, \dots\}$

2.  $\{31, 26, 21, 16, 11, \dots\}$

3.  $\{3888, -648, 108, -18, 3, \dots\}$

4.  $\left\{-2, -\frac{7}{3}, -\frac{8}{3}, -3, \dots\right\}$

5.  $\{-14, -28, -56, -112, \dots\}$

6.  $\{0, -1, -3, -6, -10, \dots\}$

7.  $\{-45, -38, -31, -24, \dots\}$

8.  $\left\{\frac{5}{9}, \frac{5}{6}, \frac{5}{4}, \frac{15}{8}, \frac{45}{16}, \dots\right\}$

9.  $\left\{\frac{1}{32}, -\frac{1}{8}, \frac{1}{2}, -2, 8, \dots\right\}$

**Determine whether the series is arithmetic or geometric, then find the indicated sum.**

10.  $\{-6561 + 2187 - 729 + 243 + \dots\}; S_9$

11.  $\left\{-\frac{3}{2} + 1 + \frac{7}{2} + 6 + \frac{17}{2} + \dots\right\}; S_{24}$

12.  $\sum_{p=1}^{34} (19 - 2p)$

13.  $\sum_{n=1}^7 -\frac{1}{8} \cdot 4^{n-1}$

$$14. \sum_{x=2}^{10} 640 \cdot \left(\frac{1}{2}\right)^{x-1}$$

$$15. \sum_{k=1}^{29} (5k - 42)$$

$$16. \sum_{m=1}^{\infty} 250 \cdot \left(\frac{1}{5}\right)^{m-1}$$

$$17. \sum_{y=1}^{\infty} -2 \cdot \left(-\frac{2}{3}\right)^{y-1}$$

**Solve each problem.**

18. The football team is running a jackpot in which the prize will start at \$2500 for the first game, and then increase \$500 each game if there is no winner. If there are no winners, find the value of the jackpot at the end of their 16-game season.

19. The value of Casey's home for the first three years after purchasing is given in the table below. If this pattern continues, find the approximate value of his home after 30 years.

| Year | Value     |
|------|-----------|
| 1    | \$185,400 |
| 2    | \$190,962 |
| 3    | \$196,691 |

20. The construction crew is pounding a stake into the ground. On the first strike, it goes 24 centimeters into the ground, 18 centimeters further on the second strike, then 13.5 centimeters further on the third strike. If this pattern continues, find the total distance the stake was driven after seven strikes.

21. The table below shows the amount of snowfall for the first three days of January in the town of Orchard Hills. If this pattern continues, find the total amount of snowfall for the month of January.

| Day | Inches |
|-----|--------|
| 1   | 0.2    |
| 2   | 0.5    |
| 3   | 0.8    |

|            | ARITHMETIC<br>Sequence                                                         | GEOMETRIC<br>Sequence                            | ARITHMETIC<br>Series                                 | GEOMETRIC<br>Series                                                       | INF. GEOMETRIC<br>Series                                                        |
|------------|--------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Definition |                                                                                |                                                  |                                                      |                                                                           |                                                                                 |
| Formula    |                                                                                |                                                  |                                                      |                                                                           |                                                                                 |
| Example 1  | Find $a_{28}$ :<br>$\{-5, 9, 23, 37, 51, \dots\}$                              | Find $a_{14}$ :<br>$\{-8, 16, -32, 64, \dots\}$  | Find $S_{37}$ :<br>$\{(-3) + (-7) + (-11) + \dots\}$ | Find $S_{12}$ :<br>$\{2 - 6 + 18 - 54 + \dots\}$                          | Find the sum:<br>$\{400 + 200 + 100 + \dots\}$                                  |
| Example 2  | Find $a_{23}$ :<br>$\left\{9, \frac{13}{2}, 4, \frac{3}{2}, -1, \dots\right\}$ | Find $a_7$ :<br>$\{1024, 768, 576, 432, \dots\}$ | Find the sum:<br>$\sum_{k=1}^{33} (6k - 15)$         | Find the sum:<br>$\sum_{m=2}^8 -512 \cdot \left(\frac{3}{2}\right)^{m-1}$ | Find the sum:<br>$\sum_{y=1}^{\infty} 48 \cdot \left(-\frac{1}{4}\right)^{y-1}$ |

# Unit 10 Test Study Guide

## (Sequences & Series)

Name: \_\_\_\_\_

Date: \_\_\_\_\_ Per: \_\_\_\_\_

### Topic 1: Sequences, Explicit & Recursive Formulas

|                                                                                             |                                                                            |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 1. What is a sequence? _____<br>_____                                                       |                                                                            |
| 2. Describe the difference between recursive and explicit formulas: _____<br>_____<br>_____ |                                                                            |
| <b>Given the formula, write the first six terms of each sequence.</b>                       |                                                                            |
| 3. $a_1 = -1$ ; $a_n = 2a_{n-1} - 5$ (for $n \geq 2$ )                                      | 4. $a_1 = 1$ , $a_2 = 2$ ; $a_n = (a_{n-1})^2 - a_{n-2}$ (for $n \geq 3$ ) |
| 5. $a_n = 3n - 8$                                                                           | 6. $a_n = \frac{1}{4} \cdot (-2)^{n-1}$                                    |
| 7. $a_n = \frac{n+4}{n+2}$                                                                  | 8. $a_n = n^3 - n^2$                                                       |

### Topic 2: Expanding & Evaluating Series

|                                                                             |                      |
|-----------------------------------------------------------------------------|----------------------|
| 9. What is a series? _____                                                  |                      |
| <b>Write each sequence as a series, then find <math>S_8</math>.</b>         |                      |
| 10. $a_1 = 2$ , $a_2 = 3$ ; $a_n = a_{n-1} \cdot a_{n-2}$ (for $n \geq 3$ ) | 11. $a_n = n^2 + 2n$ |
| <b>Expand and evaluate each series.</b>                                     |                      |
| 12. $\sum_{k=1}^{16} (k-5)^2$                                               |                      |
| 13. $\sum_{m=3}^{14} (m^3 - 4m)$                                            |                      |

### Topic 3: Arithmetic vs. Geometric Sequences

|                                                                                                                                                              | DEFINITION                                                        |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--|
| ARITHMETIC                                                                                                                                                   |                                                                   |  |
| GEOMETRIC                                                                                                                                                    |                                                                   |  |
| For each sequence: a) Determine if the sequence is arithmetic, geometric, or neither and b) Determine the common difference ( $d$ ) or common ratio ( $r$ ). |                                                                   |  |
| 14. $\{3, 12, 48, 192, 768, \dots\}$                                                                                                                         | 15. $\{16, 106, 1006, 10006, 100006, \dots\}$                     |  |
| 16. $\{2, -5, -12, -19, -26, \dots\}$                                                                                                                        | 17. $\left\{45, -15, 5, -\frac{5}{3}, \frac{5}{9}, \dots\right\}$ |  |
| 18. $\{6, 8, 11, 15, 20, \dots\}$                                                                                                                            | 19. $\{-11, -8, -5, -2, 1, \dots\}$                               |  |
| 20. $\left\{-1125, -450, -180, -72, -\frac{144}{5}, \dots\right\}$                                                                                           | 21. $\{0, 1, 1, 2, 3, 5, \dots\}$                                 |  |

### Topic 4: Writing Arithmetic & Geometric Sequence Formulas

| Arithmetic Sequence Formula:                                                                          | Geometric Sequence Formula:                                             |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Write a formula for each arithmetic sequence, then find the indicated term.                           |                                                                         |
| 22. $\{-13, -7, -1, 5, \dots\}; a_{42}$                                                               | 23. $\{34, 24, 14, 4, \dots\}; a_{30}$                                  |
| 24. $\left\{-\frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \dots\right\}; a_{21}$ | 25. $\left\{4, \frac{5}{2}, 1, -\frac{1}{2}, -2, \dots\right\}; a_{25}$ |

| Write a formula for the arithmetic sequence and find the indicated value.  |                                                             |
|----------------------------------------------------------------------------|-------------------------------------------------------------|
| 26. $a_{31} = -108$ and $d = -3$ ; find $a_1$                              | 27. $a_{14} = 15$ and $d = 9$ ; find $a_4$                  |
| 28. $a_1 = -2$ and $a_{10} = 43$ ; find $d$                                | 29. $a_1 = 10$ and $a_{21} = 2$ ; find $a_{14}$             |
| Write a formula for each geometric sequence, then find the indicated term. |                                                             |
| 30. $\{8, -16, 32, -64, \dots\}; a_{18}$                                   | 31. $\{6561, 2187, 729, 243, \dots\}; a_{11}$               |
| 32. $\left\{18, 24, 32, \frac{128}{3}, \dots\right\}; a_6$                 | 33. $\left\{-\frac{2}{5}, -2, -10, -50, \dots\right\}; a_8$ |
| Write a formula for the geometric sequence and find the indicated value.   |                                                             |
| 34. $a_5 = -10$ and $r = -\frac{1}{2}$ ; find $a_1$                        | 35. $a_1 = 2$ and $a_5 = 4802$ ; find $r$                   |
| 36. $a_3 = -18$ and $a_6 = 486$ ; find $a_1$                               | 37. $a_2 = 1500$ and $a_4 = 960$ ; find $a_6$               |

**Topic 5: Arithmetic & Geometric Series**

| Sum of an Arithmetic Series                                                          | Sum of a Geometric Series                                       | Sum of a Convergent Infinite Geometric Series |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------|
| Find the indicated sum for each series.                                              |                                                                 |                                               |
| 38. $\{6 + 1 + (-4) + (-9) + \dots\}; S_{22}$                                        | 39. $\{2 + 12 + 72 + 432 + \dots\}; S_{13}$                     |                                               |
| 40. $\left\{800 - 200 + 50 - \frac{25}{2} + \dots\right\}; S_8$                      | 41. $\{(-29) + (-27) + (-25) + (-23) + \dots\}; S_{36}$         |                                               |
| 42. $\sum_{y=1}^9 -2 \cdot (-3)^{y-1}$                                               | 43. $\sum_{n=1}^{24} (4n - 7)$                                  |                                               |
| 44. $\sum_{k=3}^{10} 64 \cdot \left(\frac{1}{2}\right)^{k-1}$                        | 45. $\sum_{m=6}^{39} (2m - 38)$                                 |                                               |
| Determine if the series is convergent or divergent. Find the sum, if possible.       |                                                                 |                                               |
| 46. $\left\{-\frac{3}{2} + \frac{3}{4} - \frac{3}{8} + \frac{3}{16} + \dots\right\}$ | 47. $\left\{24 + 6 + \frac{3}{2} + \frac{3}{8} + \dots\right\}$ |                                               |



|                                                                           |                                                                                     |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>48.</b> $\{4 + 16 + 64 + 256 + \dots\}$                                | <b>49.</b> $\{500 - 300 + 180 - 108 + \dots\}$                                      |
| <b>50.</b> $\sum_{k=1}^{\infty} 9 \cdot \left(\frac{4}{5}\right)^{k-1}$   | <b>51.</b> $\sum_{x=1}^{\infty} 10 \cdot \left(-\frac{4}{3}\right)^{x-1}$           |
| <b>52.</b> $\sum_{n=4}^{\infty} -54 \cdot \left(\frac{1}{3}\right)^{n-1}$ | <b>53.</b> $\sum_{i=5}^{\infty} -\frac{9}{2} \cdot \left(-\frac{1}{2}\right)^{i-1}$ |

### Topic 6: Applications

**54.** Bricks are stacked in a pile with 42 bricks on the bottom row and 9 bricks on the top row. Each row has three fewer bricks than the row below it. Write a formula to represent the number of bricks on each row, then determine the number of bricks on the 7<sup>th</sup> row.

**55.** A company is offering a job with a salary of \$48,000 for the first year, then a raise of 2% each year after that. Write a formula to represent the salary after each year of employment, then determine the salary after 30 years.

**56.** The table shows the number of seats in the first three rows of the concert hall. This pattern continues for a total of 12 rows. Determine if there are enough seats for a group of 500 people to attend a concert.

| Row | Seats |
|-----|-------|
| 1   | 30    |
| 2   | 32    |
| 3   | 34    |

Name: \_\_\_\_\_

Date: \_\_\_\_\_ Per: \_\_\_\_\_

## Unit 10 Test

### Sequences & Series

1. Give the first five terms of the sequence:

$$a_1 = -4; a_n = 2a_{n-1} - 1 \text{ (for } n \geq 2\text{)}$$

|       |
|-------|
| _____ |
|-------|

2. Give the first five terms of the sequence:

$$a_1 = \frac{2}{5}, a_2 = \frac{4}{5}; a_n = a_{n-2} \cdot a_{n-1} \text{ (for } n \geq 3\text{)}$$

|       |
|-------|
| _____ |
|-------|

3. Give the first five terms of the sequence:

$$a_n = n^3 - 2n$$

|       |
|-------|
| _____ |
|-------|

4. Give the first five terms of the sequence:

$$a_n = 1 - \frac{1}{n}$$

|       |
|-------|
| _____ |
|-------|

5. Which formula can be used to represent the sequence below?

$$\{2, 5, 17, 65, 257, \dots\}$$

A.  $a_n = n_3 - 3$

B.  $a_n = 4n - 3$

C.  $a_1 = 2; a_n = (a_{n-1})^3 - 3 \text{ (for } n \geq 2\text{)}$

D.  $a_1 = 2; a_n = 4a_{n-1} - 3 \text{ (for } n \geq 2\text{)}$

|  |
|--|
|  |
|--|

6. Which formula can be used to represent the sequence below?

$$\{-6, -3, 2, 9, 18, \dots\}$$

A.  $a_n = 3n - 9$

B.  $a_n = n_2 - 7$

C.  $a_1 = -6; a_n = 2a_{n-1} + 9 \text{ (for } n \geq 2\text{)}$

D.  $a_1 = -6; a_n = \frac{1}{2}a_{n-1} + 1 \text{ (for } n \geq 2\text{)}$

|  |
|--|
|  |
|--|

7. Find the sum of the first nine terms in the series below.

$$\{1 + 8 + 27 + 64 + 125 + \dots\}$$

|  |
|--|
|  |
|--|

8. Expand and evaluate the series:

$$\sum_{k=1}^{15} (5 - 2k^2)$$

|  |
|--|
|  |
|--|

**For questions 9-12, describe the sequence as arithmetic, geometric, or neither.  
For arithmetic and geometric sequences, give the common difference or common ratio.**

9.  $\{3, -6, 12, -24, -48, \dots\}$

Type of Sequence

Common  
Difference/Ratio

10.  $\left\{-\frac{11}{4}, -2, -\frac{5}{4}, -\frac{1}{2}, \frac{1}{4}, \dots\right\}$

Type of Sequence

Common  
Difference/Ratio

11.  $\{3600, 1440, 576, 230.4, \dots\}$

Type of Sequence

Common  
Difference/Ratio

12.  $\{600, -300, 100, -25, \dots\}$

Type of Sequence

Common  
Difference/Ratio

13. Identify the formula for the sequence given below.

$$\{11, 7, 3, -1, -5, \dots\}$$

A.  $a_n = 15 - 4n$

B.  $a_n = 7 - 4n$

C.  $a_n = 11 \cdot (-4)^n$

D.  $a_n = 11 \cdot (-4)^{n-1}$

14. Identify the formula for the sequence given below.

$$\{76.8, 96, 120, 150, \dots\}$$

A.  $a_n = 76.8 + \frac{4}{5}n$

C.  $a_n = 76.8 \cdot \left(\frac{4}{5}\right)^{n-1}$

B.  $a_n = 76.8 + \frac{5}{4}n$

D.  $a_n = 76.8 \cdot \left(\frac{5}{4}\right)^{n-1}$

15. Write a formula for the sequence below, then find  $a_{10}$ .

$$\{-4, -12, -36, -108, \dots\}$$

Formula

$a_{10} =$

16. Write a formula for the sequence below, then find  $a_{45}$ .

$$\left\{-2, -\frac{7}{4}, -\frac{3}{2}, -\frac{5}{4}, -1, \dots\right\}$$

Formula

$a_{45} =$

17. If the information given below represents an arithmetic sequence, find  $a_1$ .

$$a_{14} = 16 \text{ and } d = -\frac{3}{2}$$

$$a_1 =$$

18. If the information given below represents an arithmetic sequence, find  $a_{27}$ .

$$a_1 = 49 \text{ and } a_8 = 14$$

$$a_{27} =$$

19. If the information given below represents a geometric sequence, find  $a_1$ .

$$a_4 = -320 \text{ and } r = -4$$

$$a_1 =$$

20. If the information given below represents a geometric sequence, find  $a_3$ .

$$a_1 = 3645 \text{ and } a_6 = 15$$

$$a_3 =$$

21. Given the sequence below, find  $S_{19}$ .

$$\{30, 23, 16, 9, 2, \dots\}$$

- A. -573  
B. -592  
C. -627  
D. -649

22. Given the sequence below, find  $S_8$ .

$$\{3, -12, 48, -192, \dots\}$$

- A. 65535  
B. -65535  
C. 39321  
D. -39321

23. Given the sequence below, find  $S_7$ .

$$\{4374, 1458, 486, 162, \dots\}$$

- A. 6558  
B. 6736  
C. 7014  
D. 7252

24. Given the sequence below, find  $S_{25}$ .

$$\left\{1, \frac{5}{3}, \frac{7}{3}, 3, \frac{11}{3}, \dots\right\}$$

- A. 208  
B. 225  
C. 240  
D. 272

25. Find the sum:  $\sum_{n=1}^{34} (-17 + 8n)$

- A. 3806  
B. 4074  
C. 4182  
D. 4468

26. Find the sum:  $\sum_{x=2}^7 16 \cdot \left(\frac{5}{2}\right)^{x-1}$

- A. 6483.75  
B. 6499.75  
C. 16249.375  
D. 2593.5

27. Which series do not have a sum? Check all that apply.

|                                                                                                            |                                                                                                           |                                                                                          |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <input type="checkbox"/> $\{343 + 98 + 28 + 8 + \dots\}$                                                   | <input type="checkbox"/> $\left\{36 + 9 + \frac{9}{4} + \frac{9}{16} + \dots\right\}$                     | <input type="checkbox"/> $\sum_{k=1}^{\infty} -120 \cdot \left(\frac{4}{3}\right)^{k-1}$ |
| <input type="checkbox"/> $\left\{\frac{5}{3} - \frac{10}{3} + \frac{20}{3} - \frac{40}{3} + \dots\right\}$ | <input type="checkbox"/> $\left\{\frac{1}{2} - \frac{3}{4} + \frac{9}{8} - \frac{27}{16} + \dots\right\}$ | <input type="checkbox"/> $\sum_{n=1}^{\infty} 180 \cdot \left(-\frac{1}{5}\right)^{n-1}$ |

28. Find the sum:  $\left\{18 - 12 + 8 - \frac{16}{3} + \dots\right\}$

- A. 10  
B. 10.8  
C. 48  
D. 54

29. Find the sum:  $\sum_{m=1}^{\infty} -32 \cdot \left(\frac{1}{4}\right)^{m-1}$

- A.  $\frac{128}{5}$   
B.  $-\frac{128}{5}$   
C.  $\frac{128}{3}$   
D.  $-\frac{128}{3}$

30. Mark got a new job with a starting salary of \$32,000. If he gets a yearly raise of 2%, find Mark's salary after 25 years.

- 31.** An auditorium has 21 rows of seats. The first row has 18 seats, and each succeeding row has two more seats than the previous row. How many seats are in the auditorium?

- 32.** Jalen is in a 24 hour dance-a-thon as part of a fundraiser. He raised \$50 in the first hour of the competition, then another \$60 in the second hour, then another \$72 in the third hour. If this pattern continues, what is the total amount of money he will raise if he stays for the entire competition?

|              |              |
|--------------|--------------|
| Name: _____  | Date: _____  |
| Topic: _____ | Class: _____ |

| Main Ideas/Questions      | Notes/Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |   |    |    |   |   |       |   |   |   |    |    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|----|----|---|---|-------|---|---|---|----|----|
| SEQUENCE                  | A set of numbers with a particular order or pattern. Each number is called a "term."                                                                                                                                                                                                                                                                                                                                                                                           |     |   |    |    |   |   |       |   |   |   |    |    |
| FINITE<br>Sequence        | A sequence with a limited number of terms.<br>Example: {1, 2, 3, 4, 5}                                                                                                                                                                                                                                                                                                                                                                                                         |     |   |    |    |   |   |       |   |   |   |    |    |
| INFINITE<br>Sequence      | A sequence with unlimited terms; not ending<br>Example: {0, 1, 2, 3, 4, 5, ...} ← set of whole #'s                                                                                                                                                                                                                                                                                                                                                                             |     |   |    |    |   |   |       |   |   |   |    |    |
| TERM<br>Notation          | <ul style="list-style-type: none"><li>The first term in a sequence is denoted <math>a_1</math>.</li><li>Each subsequent term is denoted <math>a_n</math>, where <math>n</math> is the term number in the sequence.</li></ul> Example: Given {1, 5, 9, 13, 17, ...}, identify the following term values:<br>$a_1$ : 1 $a_4$ : 13 $a_9$ : 33 $a_{12}$ : 45                                                                                                                       |     |   |    |    |   |   |       |   |   |   |    |    |
| SEQUENCES<br>as Functions | Since each term value is paired with exactly one term number, a sequence is a function with the following properties:<br><br>The domain is the set of term numbers.<br><table><tr><td><math>n</math></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td><math>a_n</math></td><td>1</td><td>5</td><td>9</td><td>13</td><td>17</td></tr></table><br>The range is the set of term values.<br><br>In an infinite set, the domain is the the set of natural numbers | $n$ | 1 | 2  | 3  | 4 | 5 | $a_n$ | 1 | 5 | 9 | 13 | 17 |
| $n$                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2   | 3 | 4  | 5  |   |   |       |   |   |   |    |    |
| $a_n$                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5   | 9 | 13 | 17 |   |   |       |   |   |   |    |    |
| RECURSIVE<br>Formulas     | A rule in which one or more previous terms are used to generate the next term.<br>The Fibonacci Sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...<br>In this sequence, $a_1 = 0$ , $a_2 = 1$ , then for each subsequent term,<br>$a_n = a_{n-1} + a_{n-2}$                                                                                                                                                                                                                        |     |   |    |    |   |   |       |   |   |   |    |    |
| EXAMPLES                  | Find the first 5 terms of each sequence.<br>1. $a_1 = 14$ ; $a_n = a_{n-1} + 9$ (for $n \geq 2$ )<br>14, 23, 32, 41, 50<br>2. $a_1 = 6$ ; $a_n = 2a_{n-1} - 5$ (for $n \geq 2$ )<br>6, 7, 9, 13, 21                                                                                                                                                                                                                                                                            |     |   |    |    |   |   |       |   |   |   |    |    |

|                                    |                                                                                                                                                                           |                                                                                                                                                             |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>3. <math>a_1 = 354</math>; <math>a_n = \frac{a_{n-1} + 6}{3}</math> (for <math>n \geq 2</math>)</p> <p>354, 120, 42, 16, <math>\frac{22}{3}</math></p>                 | <p>4. <math>a_1 = 3</math>, <math>a_2 = 4</math>; <math>a_n = a_{n-1} \cdot a_{n-2}</math> (for <math>n \geq 3</math>)</p> <p>3, 4, 12, 48, 576</p>         |
|                                    | Write a rule for each sequence. Then give the next 3 terms.                                                                                                               |                                                                                                                                                             |
|                                    | <p>5. {4, 11, 32, 95, 284, ...}</p> <p><math>a_1 = 4</math>; <math>a_n = 3 \cdot a_{n-1} - 1</math></p> <p>851, 2552, 7655</p>                                            | <p>6. {100, 60, 40, 30, 25, ...}</p> <p><math>a_1 = 100</math>; <math>a_n = \frac{1}{2} a_{n-1} + 10</math></p> <p>22.5, 21.25, 20.625</p>                  |
| <b>EXPLICIT</b><br><i>Formulas</i> | A rule in which the $n^{\text{th}}$ term is defined as a function of $n$ . * Previous terms are unnecessary. *                                                            |                                                                                                                                                             |
| <b>EXAMPLES</b>                    | Find the first 5 terms of each sequence.                                                                                                                                  |                                                                                                                                                             |
|                                    | <p>7. <math>a_n = 7(n - 3)</math></p> <p>-14, -7, 0, 7, 14</p>                                                                                                            | <p>8. <math>a_n = \left(\frac{1}{3}\right)^{n+1}</math></p> <p><math>\frac{1}{9}, \frac{1}{27}, \frac{1}{81}, \frac{1}{243}, \frac{1}{729}</math></p>       |
|                                    | <p>9. <math>a_n = -4n^2</math></p> <p>-4, -16, -36, -64, -100</p>                                                                                                         | <p>10. <math>a_n = 2n(n + 5)</math></p> <p>12, 28, 48, 72, 100</p>                                                                                          |
|                                    | <p>11. <math>a_n = (-2)^{n-1}</math></p> <p>1, -2, 4, -8, 16</p>                                                                                                          | <p>12. <math>a_n = (n - 4)^2</math></p> <p>9, 4, 1, 0, 1</p>                                                                                                |
|                                    | Write a rule for each sequence. Then give the next 3 terms.                                                                                                               |                                                                                                                                                             |
|                                    | <p>13. {-7, -9, -11, -13, -15, ...}</p> <p><math>a_n = -2n - 5</math>;</p> <p>-17, -19, -21</p>                                                                           | <p>14. {3, 6, 11, 18, 27, ...}</p> <p><math>a_n = n^2 + 2</math>;</p> <p>38, 51, 66</p>                                                                     |
|                                    | <p>15. <math>\left\{2, \frac{5}{2}, 3, \frac{7}{5}, 4, \dots\right\}</math></p> <p><math>a_n = \frac{n+3}{2}</math>;</p> <p><math>\frac{9}{2}, 5, \frac{11}{2}</math></p> | <p>16. <math>\{1, \sqrt{2}, \sqrt{3}, 2, \sqrt{5}, \dots\}</math></p> <p><math>a_n = \sqrt{n}</math>;</p> <p><math>\sqrt{6}, \sqrt{7}, 2\sqrt{2}</math></p> |



|                      |                                                                                                                                                                                                                                                                                                                              |                                                                                                             |                   |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------|
| Name:                |                                                                                                                                                                                                                                                                                                                              | Date:                                                                                                       |                   |
| Topic:               |                                                                                                                                                                                                                                                                                                                              | Class:                                                                                                      |                   |
| Main Ideas/Questions |                                                                                                                                                                                                                                                                                                                              | Notes/Examples                                                                                              |                   |
| SERIES               | The sum of the terms in a sequence.                                                                                                                                                                                                                                                                                          |                                                                                                             |                   |
|                      | Sequence                                                                                                                                                                                                                                                                                                                     | {1, 2, 3, 4}                                                                                                | {3, 6, 9, 12,...} |
|                      | Series                                                                                                                                                                                                                                                                                                                       | 10                                                                                                          | $\infty$          |
| PARTIAL SUMS         | The sum of a specified number of terms.                                                                                                                                                                                                                                                                                      |                                                                                                             |                   |
|                      | Find the partial sum for each given sequence.                                                                                                                                                                                                                                                                                |                                                                                                             |                   |
|                      | 1. {1, 2, 3, 4, 5,...}; find $S_5$<br>6, 7, 8, 9<br>$= 45$                                                                                                                                                                                                                                                                   | 2. {4, 7, 13, 25, 49,...}; find $S_4$<br>$= 49$                                                             |                   |
| SUMMATION            | 3. {1, 4, 9, 16, 25,...}; find $S_6$<br>36<br>$= 91$                                                                                                                                                                                                                                                                         | 4. {1, $\frac{1}{2}$ , $\frac{1}{3}$ , $\frac{1}{4}$ , $\frac{1}{5}$ , ...}; find $S_5$<br>$= \frac{11}{6}$ |                   |
|                      | A way to represent a series using the greek letter $\Sigma$ to denote the sum.<br>$\uparrow$ sigma                                                                                                                                                                                                                           |                                                                                                             |                   |
| Notation             | <div>last term # <math>\rightarrow</math> 5</div> <div>first term # <math>\rightarrow</math> n=1</div> <div><math>\sum_{n=1}^5 2n</math></div> <div>formula for the sequence</div> <div>Find the sum of the series above:<br/><math>2(1) + 2(2) + 2(3) + 2(4) + 2(5)</math><br/><math>= 2 + 4 + 6 + 8 + 10 = 30</math></div> |                                                                                                             |                   |
| EXAMPLES             | Expand each series and evaluate.                                                                                                                                                                                                                                                                                             |                                                                                                             |                   |
|                      | 5. $\sum_{n=1}^{12} (n-1)$<br>$0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 = 66$                                                                                                                                                                                                                                         |                                                                                                             |                   |
|                      | 6. $\sum_{n=1}^7 (-3n)$<br>$-3 + (-6) + (-9) + (-12) + (-15) + (-18) + (-21) = -84$                                                                                                                                                                                                                                          |                                                                                                             |                   |

$$7. \sum_{k=1}^4 (5k + 2)$$

$$= 7 + 12 + 17 + 22 = \boxed{58}$$

$$8. \sum_{a=2}^9 (2a - 7)$$

$$= (-3) + (-1) + 1 + 3 + 5 + 7 + 9 + 11 = \boxed{32}$$

$$9. \sum_{y=1}^5 (y^2 + y)$$

$$= (1+1) + (4+2) + (9+3) + (16+4) + (25+5)$$

$$= 2 + 6 + 12 + 20 + 30 = \boxed{70}$$

$$10. \sum_{c=2}^8 \left( \frac{c^3}{2} \right)$$

$$= \frac{8}{2} + \frac{27}{2} + \frac{64}{2} + \frac{125}{2} + \frac{216}{2} + \frac{343}{2} + \frac{512}{2}$$

$$= 4 + 13.5 + 32 + 62.5 + 108 + 171.5 + 256 = \boxed{647.5}$$

$$11. \sum_{m=2}^6 (m^2 - 1)$$

$$= (4-1) + (9-1) + (16-1) + (25-1) + (36-1)$$

$$= 3 + 8 + 15 + 24 + 35 = \boxed{85}$$

$$12. 3 \cdot \sum_{p=4}^8 (4 - p)^2$$

$$= 3 \left[ (0)^2 + (-1)^2 + (-2)^2 + (-3)^2 + (-4)^2 \right]$$

$$= 3 \left[ 0 + 1 + 4 + 9 + 16 \right] = 3 \left[ 30 \right] = \boxed{90}$$

$$13. \frac{1}{2} \cdot \sum_{x=2}^4 (x^2 - 5)$$

$$= \frac{1}{2} \left[ (4-5) + (9-5) + (16-5) \right]$$

$$= \frac{1}{2} \left[ -1 + 4 + 11 \right] = \frac{1}{2} \left[ 14 \right] = \boxed{7}$$

Name: \_\_\_\_\_

Unit 10: Sequences &amp; Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 1: Intro to Sequences &amp; Series

\*\* This is a 2-page document! \*\*

Find the first five terms of each sequence.

1.  $a_1 = 9; a_n = a_{n-1} - 2$  (for  $n \geq 2$ )

9, 7, 5, 3, 1

2.  $a_1 = -4; a_n = -3a_{n-1} + 1$  (for  $n \geq 2$ )

-4, 13, -38, 115, -344

3.  $a_1 = 120; a_n = \frac{1}{2}a_{n-1} + 8$  (for  $n \geq 2$ )

120, 68, 42, 29, 22.5

4.  $a_1 = 2, a_2 = 5; a_n = a_{n-1} \cdot a_{n-2} - 4$  (for  $n \geq 3$ )

2, 5, 6, 26, 152

Write a possible recursive rule, then find  $a_7$ .

5.  $\{7, 12, 17, 22, 27, \dots\}$

$a_1 = 7; a_n = a_{n-1} + 5$  ( $n \geq 2$ )

$a_7 = 37$

6.  $\{2, 3, 8, 19, 46, \dots\}$

$a_1 = 2, a_2 = 3; a_n = a_{n-2} + 2 \cdot a_{n-1}$  ( $n \geq 3$ )

$a_7 = 268$

Find the first five terms of each sequence.

7.  $a_n = -2(n-7)$

12, 10, 8, 6, 4

8.  $a_n = \left(\frac{1}{2}\right)^{n+3}$

 $\frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128}, \frac{1}{256}$ 

9.  $a_n = 4n^2$

4, 16, 36, 64, 100

10.  $a_n = (n+1)^3$

8, 27, 64, 125, 216

Write a possible explicit rule, then find  $a_{10}$ .

11.  $\{-7, -6, -5, -4, -3, \dots\}$

$a_n = n - 8; a_{10} = 2$

12.  $\{5, 7, 9, 11, 13, \dots\}$

$a_n = 2n + 3; a_{10} = 23$

13.  $\{0, 7, 26, 63, 124, \dots\}$

$a_n = n^3 - 1; a_{10} = 999$

14.  $\left\{\frac{1}{3}, \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{5}{7}, \dots\right\}$

$a_n = \frac{n}{n+2}; a_{10} = \frac{5}{6}$

Find the partial sum for each sequence.

15.  $\{38, 31, 24, 17, \dots\}; S_{12}$

10, 3, -4, -11, -18, -25, -32, -39

$S_{12} = -6$

16.  $\{1, 8, 27, 64, 125, \dots\}; S_7$

216, 343

$S_7 = 784$

Expand each series and evaluate.

$$17. \sum_{n=1}^{14} (n-6)$$

$$= (-5) + (-4) + (-3) + (-2) + (-1) + 0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8$$

$$= 21$$

$$18. \sum_{p=1}^9 (4p)$$

$$= 4 + 8 + 12 + 16 + 20 + 24 + 28 + 32 + 36$$

$$= 180$$

$$19. \sum_{m=1}^5 (-2m+7)$$

$$= (2+7) + (-4+7) + (-6+7) + (-8+7) + (-10+7)$$

$$= 5 + 3 + 1 + (-1) + (-3) = 5$$

$$20. \sum_{y=1}^8 (3y^2 - 2y)$$

$$= (3-2) + (12-4) + (27-6) + (48-8) + (75-10) + (108-12) + (147-14) + (192-16)$$

$$= 1 + 8 + 21 + 40 + 65 + 96 + 133 + 176 = 540$$

$$21. \sum_{c=3}^{10} \left( \frac{c^2+1}{2} \right)$$

$$= \frac{10}{2} + \frac{17}{2} + \frac{26}{2} + \frac{37}{2} + \frac{50}{2} + \frac{65}{2} + \frac{82}{2} + \frac{101}{2}$$

$$= 5 + 8.5 + 13 + 18.5 + 25 + 32.5 + 41 + 50.5 = 194$$

$$22. \sum_{n=2}^7 \left( \frac{n-2}{n+2} \right)$$

$$= \frac{0}{4} + \frac{1}{5} + \frac{2}{6} + \frac{3}{7} + \frac{4}{8} + \frac{5}{9}$$

$$= \frac{1271}{630}$$

$$23. 2 \cdot \sum_{a=5}^{10} (a+2)^2$$

$$= 2 \cdot [49 + 64 + 81 + 100 + 121 + 144 + 169 + 196 + 225]$$

$$= 2 \cdot [1149] = 2298$$

$$24. \frac{1}{4} \cdot \sum_{r=3}^{11} (15-2r)$$

$$= \frac{1}{4} \cdot [9 + 7 + 5 + 3 + 1 + (-1) + (-3) + (-5) + (-7)]$$

$$= \frac{1}{4} \cdot [9] = \frac{9}{4}$$

|        |        |
|--------|--------|
| Name:  | Date:  |
| Topic: | Class: |

| Main Ideas/Questions                                   | Notes/Examples                                                                                                                                                                                                                 |                                                                                                                                                                                         |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ARITHMETIC</b><br><i>Sequence</i>                   | A sequence in which the difference between any two terms is constant.                                                                                                                                                          |                                                                                                                                                                                         |
| <i>Common</i><br><b>DIFFERENCE</b>                     | The numerical difference between two terms. (variable: $d$ )                                                                                                                                                                   |                                                                                                                                                                                         |
| <b>EXAMPLES</b>                                        | Determine whether the sequence is arithmetic. If yes, identify the common difference and give the next three terms.                                                                                                            |                                                                                                                                                                                         |
|                                                        | 1. $\{5, 8, 11, 14, 17, \dots\}$<br>yes; $d = 3$<br>20, 23, 26                                                                                                                                                                 | 2. $\{23, 16, 9, 2, -5, \dots\}$<br>yes; $d = -7$<br>-12, -19, -26                                                                                                                      |
|                                                        | 3. $\{1, 2, 3, 5, 8, \dots\}$<br>No                                                                                                                                                                                            | 4. $\{-6, -8, -10, -12, -14, \dots\}$<br>yes; $d = -2$<br>-16, -18, -20                                                                                                                 |
|                                                        | 5. $\{-20, -10, -5, 0, 5, \dots\}$<br>No                                                                                                                                                                                       | 6. $\{-35, -23, -11, 1, 13, \dots\}$<br>yes; $d = 12$<br>25, 37, 49                                                                                                                     |
| <b>ARITHMETIC</b><br><b>SEQUENCE</b><br><i>Formula</i> | The $n^{\text{th}}$ term of any arithmetic sequence can be found using the formula:<br><div><math display="block">a_n = d(n-1) + a_1</math></div> where $a_1$ is the <u>first term</u> and $d$ is the <u>common difference</u> |                                                                                                                                                                                         |
| <b>EXAMPLES</b>                                        | Write a rule for each sequence, then find the indicated term.                                                                                                                                                                  |                                                                                                                                                                                         |
|                                                        | 7. $\{-5, -1, 3, 7, \dots\}; a_{24}$ $a_1 = -5, d = 4$<br>$a_n = 4(n-1) + (-5)$<br>$a_n = 4n - 4 - 5$<br>$a_n = \boxed{4n - 9}$<br>$a_{24} = 4(24) - 9 = \boxed{87}$                                                           | 8. $\{20, 11, 2, -7, -16, \dots\}; a_{15}$ $a_1 = 20, d = -9$<br>$a_n = -9(n-1) + 20$<br>$a_n = -9n + 9 + 20$<br>$a_n = \boxed{-9n + 29}$<br>$a_{15} = -9(15) + 29 = \boxed{-106}$      |
|                                                        | 9. $\{-6, -9, -12, -15, \dots\}; a_{38}$ $a_1 = -6, d = -3$<br>$a_n = -3(n-1) - 6$<br>$a_n = -3n + 3 - 6$<br>$a_n = \boxed{-3n - 3}$<br>$a_{38} = -3(38) - 3 = \boxed{-117}$                                                   | 10. $\{8, 9.5, 11, 12.5, \dots\}; a_{48}$ $a_1 = 8, d = 1.5$<br>$a_n = 1.5(n-1) + 8$<br>$a_n = 1.5n - 1.5 + 8$<br>$a_n = \boxed{1.5n + 6.5}$<br>$a_{48} = 1.5(48) + 6.5 = \boxed{78.5}$ |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a_n = d(n-1) + a_1$                | <p>If the given terms are part of an arithmetic sequence, write a rule for the sequence, then find <math>a_{12}</math>.</p> <div> <div> <p>11. <math>a_1 = 3</math> and <math>a_3 = -7</math></p> <math display="block">-7 = d(3-1) + 3</math> <math display="block">-7 = 3d + 3 - d</math> <math display="block">-10 = 2d</math> <math display="block">-5 = d</math> <hr/> <math display="block">a_n = -5(n-1) + 3</math> <math display="block">a_n = -5n + 8</math> <hr/> <math display="block">a_{12} = -5(12) + 8 = \boxed{-52}</math> </div> <div> <p>12. <math>a_1 = -17</math> and <math>a_{24} = 6</math></p> <math display="block">6 = d(24-1) - 17</math> <math display="block">6 = 24d - d - 17</math> <math display="block">23 = 23d</math> <math display="block">1 = d</math> <hr/> <math display="block">a_n = 1(n-1) - 17</math> <math display="block">a_n = n - 18</math> <hr/> <math display="block">a_{12} = 12 - 18 = \boxed{-6}</math> </div> </div>                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>ARITHMETIC</b><br/>Series</p> | <p>Recall that a series is the <u>Sum</u> of a sequence. When finding the sum of an <b>arithmetic series</b>, use the following formula:</p> <div style="border: 1px solid black; padding: 10px; width: fit-content; margin: 10px auto;"> <math display="block">S_n = n \left( \frac{a_1 + a_n}{2} \right)</math> </div> <p>where <math>n</math> is the <u>number of terms</u>,<br/> <math>a_1</math> is the <u>first term</u>, and <math>a_n</math> is the <u>last term</u>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>EXAMPLES</b></p>              | <p>Find the indicated sum for each arithmetic series.</p> <div> <div> <p>13. <math>\{2 + 4 + 6 + 8 + \dots\}; S_{10}</math></p> <math display="block">a_{10} = 2(10-1) + 2</math> <math display="block">a_{10} = 20</math> <hr/> <math display="block">S_{10} = 10 \left( \frac{2+20}{2} \right)</math> <math display="block">= 10(11)</math> <math display="block">\boxed{= 110}</math> </div> <div> <p>14. <math>\{(-6) + (-19) + (-32) + (-45) + \dots\}; S_8</math></p> <math display="block">a_8 = -13(8-1) - 6</math> <math display="block">a_8 = -97</math> <hr/> <math display="block">S_8 = 8 \left( \frac{-6-97}{2} \right)</math> <math display="block">= 8(-51.5)</math> <math display="block">\boxed{= -412}</math> </div> </div> <div> <div> <p>15. <math>\sum_{x=1}^{12} (6x - 4)</math></p> <math display="block">a_1 = 2, a_n = 68, n = 12</math> <math display="block">S_{12} = 12 \left( \frac{2+68}{2} \right)</math> <math display="block">= 12(35)</math> <math display="block">\boxed{= 420}</math> </div> <div> <p>16. <math>\sum_{m=8}^{17} (-2m + 3)</math></p> <math display="block">a_1 = -13, a_n = -31, n = 10</math> <math display="block">S_{10} = 10 \left( \frac{-13-31}{2} \right)</math> <math display="block">= 10(-22)</math> <math display="block">\boxed{= -220}</math> </div> </div> |

Name: \_\_\_\_\_

Unit 10: Sequences &amp; Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 2: Arithmetic Sequences &amp; Series

**\*\* This is a 2-page document! \*\*****Determine whether the sequence is arithmetic. If yes, identify the common difference and give the next three terms.**

1.  $\{-3, -12, -48, -192, \dots\}$

No

2.  $\{-18, -27, -36, -45, \dots\}$

Yes;  $d = -9$   
-54, -63, -72

3.  $\{4, 11, 18, 25, \dots\}$

Yes;  $d = 7$   
32, 39, 46

4.  $\{16, 8, 4, 1, \dots\}$

No

5.  $\{-9, -33, -57, -81, \dots\}$

Yes;  $d = -24$   
-105, -129, -153

6.  $\{\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots\}$

Yes;  $d = \frac{1}{3}$   
 $\frac{11}{6}, \frac{13}{6}, \frac{5}{2}$ **Write a rule for each sequence, then find the indicated term.**

7.  $\{12, 27, 42, 57, \dots\}; a_{18}$

$a_1 = 12, d = 15$

$a_n = 15(n-1) + 12$

$a_n = 15n - 3$

$a_{18} = 15(18) - 3 = \boxed{267}$

8.  $\{29, 9, -11, -31, \dots\}; a_{32}$

$a_1 = 29, d = -20$

$a_n = -20(n-1) + 29$

$a_n = -20n + 49$

$a_{32} = -20(32) + 49 = \boxed{-591}$

9.  $\{-40, -44, -48, -52, \dots\}; a_{49}$

$a_1 = -40, d = -4$

$a_n = -4(n-1) - 40$

$a_n = -4n - 36$

$a_{49} = -4(49) - 36 = \boxed{-232}$

10.  $\{50, 39, 28, 17, \dots\}; a_{54}$

$a_1 = 50, d = -11$

$a_n = -11(n-1) + 50$

$a_n = -11n + 61$

$a_{54} = -11(54) + 61 = \boxed{-533}$

**11. Karen's New Year's resolution is to work out each day on her elliptical. The number of calories she has burned each day since she started is shown below. Write a rule for the sequence, and find the number of calories that Karen will burn on the 30<sup>th</sup> day.**

$\{349, 357, 365, 373, 381, \dots\}$

$a_1 = 349$

$d = 8$

$a_n = 8(n-1) + 349$

$a_n = 8n + 341$

$a_{30} = 8(30) + 341 = \boxed{581}$

If the given terms are part of an arithmetic sequence, write a rule for the sequence, then find  $a_9$ .

12.  $a_1 = 11$  and  $a_5 = 27$

$$27 = d(5-1) + 11$$

$$27 = 5d - d + 11$$

$$16 = 4d$$

$$4 = d$$

$$a_n = 4(n-1) + 11$$

$$a_n = 4n + 7$$

$$a_9 = 4(9) + 7$$

$$= 43$$

13.  $a_1 = 3$  and  $a_{14} = -23$

$$-23 = d(14-1) + 3$$

$$-23 = 13d + 3$$

$$-26 = 13d$$

$$-2 = d$$

$$a_n = -2(n-1) + 3$$

$$a_n = -2n + 5$$

$$a_9 = -2(9) + 5$$

$$= -13$$

14.  $a_1 = -5$  and  $a_7 = 91$

$$91 = d(7-1) - 5$$

$$91 = 6d - 5$$

$$96 = 6d$$

$$16 = d$$

$$a_n = 16(n-1) - 5$$

$$a_n = 16n - 21$$

$$a_9 = 16(9) - 21$$

$$= 123$$

15.  $a_1 = \frac{1}{2}$  and  $a_{23} = 6$

$$6 = d(23-1) + \frac{1}{2}$$

$$6 = 22d + \frac{1}{2}$$

$$5.5 = 22d$$

$$0.25 = d$$

$$a_n = 0.25(n-1) + \frac{1}{2}$$

$$a_n = 0.25n + 0.25$$

$$a_9 = 0.25(9) + 0.25$$

$$= 2.5$$

Find the indicated sum for each arithmetic series.

16.  $\{4 + 6 + 8 + 10 + \dots\}; S_{21}$

$$a_n = 2(21-1) + 4$$

$$= 2(20) + 4$$

$$= 44$$

$$S_{21} = 21 \left( \frac{4 + 44}{2} \right)$$

$$= 21(24) = 504$$

17.  $\{24 + 31 + 38 + 45 + \dots\}; S_9$

$$a_n = 7(9-1) + 24$$

$$= 7(8) + 24$$

$$= 80$$

$$S_9 = 9 \left( \frac{24 + 80}{2} \right)$$

$$= 9(52) = 468$$

18.  $\{15 + 4 + (-7) + (-18) + \dots\}; S_{12}$

$$a_n = -11(12-1) + 15$$

$$= -11(11) + 15$$

$$= -106$$

$$S_{12} = 12 \left( \frac{15 - 106}{2} \right)$$

$$= 12(-45.5) = -546$$

19.  $\sum_{k=1}^{25} (5k + 1)$

$$a_1 = 6, a_n = 126, n = 25$$

$$S_{25} = 25 \left( \frac{6 + 126}{2} \right)$$

$$= 25(66)$$

$$= 1650$$

20.  $\sum_{n=9}^{14} (17 - 6n)$

$$a_1 = -37, a_n = -67, n = 6$$

$$S_{14} = 6 \left( \frac{-37 - 67}{2} \right)$$

$$= 6(-52)$$

$$= -312$$

21.  $\sum_{x=11}^{35} (3x - 12)$

$$a_1 = 21, a_n = 93, n = 25$$

$$S_{25} = 25 \left( \frac{21 + 93}{2} \right)$$

$$= 25(57)$$

$$= 1425$$



Name: \_\_\_\_\_

## Algebra 2

Date: \_\_\_\_\_ Per: \_\_\_\_\_

## Unit 10: Sequences &amp; Series

## Quiz 10-1: Intro to Seq. &amp; Ser., Arithmetic Seq. &amp; Ser.

## Part I: Intro to Sequences &amp; Series

Give the first five terms of each sequence.

1.  $a_1 = -4; a_n = a_{n-1} - 6$  (for  $n \geq 2$ )

1. -4, -10, -16, -22, -28

2.  $a_1 = 3, a_2 = 8; a_n = 2a_{n-2} + a_{n-1}$  (for  $n \geq 3$ )

2. 3, 8, 14, 30, 58

3.  $a_n = 15 - 2n$

3. 13, 11, 9, 7, 5

4.  $a_n = n^3 - 20$

4. -19, -12, 7, 44, 105

5. Write a recursive formula to represent the sequence, then find  $a_7$ .

$\{4, 11, 32, 95, 284, \dots\}$

851, 2552

5.  $a_n = 3 \cdot a_{n-1} - 1$  ( $n \geq 2$ )

$a_7 =$  2552

6. Write an explicit formula to represent the sequence, then find  $a_{12}$ .

$\{8, 11, 16, 23, 32, \dots\}$

$$\begin{aligned} a_{12} &= 12^2 + 7 \\ &= 144 + 7 \\ &= 151 \end{aligned}$$

6.  $a_n = n^2 + 7$

$a_{12} =$  151

Find the partial sum for each sequence.

7.  $\{2, 3, 5, 8, 13, \dots\}; S_8$

21, 34, 55

7. 141

8.  $\{0, -1, -3, -6, -10, \dots\}; S_{14}$

-15, -21, -28, -36, -45, -55, -66, -78, -91

8. -455

Expand and evaluate the series.

9.  $\sum_{a=1}^{10} (a-2)^2 = (-1)^2 + 0^2 + 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 = 1 + 0 + 1 + 4 + 9 + 16 + 25 + 36 + 49 + 64 =$  205

## Part II: Arithmetic Sequences & Series

Determine whether the sequence is arithmetic. If yes, identify the common difference.

10.  $\{1, 2, 3, 5, 8, \dots\}$

10. No

11.  $\{17, 9, 1, -7, -15, \dots\}$

11. Yes;  $d = -8$

12.  $\left\{\frac{4}{5}, \frac{6}{5}, \frac{8}{5}, 2, \frac{12}{5}, \dots\right\}$

12. Yes;  $d = \frac{2}{5}$

Write a rule for each sequence, then find the indicated term.

13.  $\{-7, -4, -1, 2, \dots\}; a_{38}$

$$a_n = 3(n-1) - 7$$

$$= 3n - 10$$

$$a_{38} = 3(38) - 10$$

$$= 104$$

13.  $a_n = 3n - 10$

$a_{38} =$  104

14.  $\{12, 10.5, 9, 7.5, \dots\}; a_{55}$

$$a_n = -1.5(n-1) + 12$$

$$= -1.5n + 13.5$$

$$a_{55} = -1.5(55) + 13.5$$

$$= -69$$

14.  $a_n = -1.5n + 13.5$

$a_{57} =$  -69

15. The terms below are part of an arithmetic sequence.

Write the rule, then find  $a_9$ .

$a_1 = -27$  and  $a_{18} = 41$

$$41 = d(18-1) - 27$$

$$68 = 17d$$

$$4 = d$$

$$a_n = 4(n-1) - 27$$

$$= 4n - 31$$

$$a_9 = 4(9) - 31 = 5$$

15.  $a_n = 4n - 31$

$a_9 =$  5

Find the indicated sum for each arithmetic series.

16.  $\{(-2) + (-7) + (-12) + (-17) + \dots\}; S_{27}$

$$a_n = -5(n-1) - 2$$

$$= -5n + 3$$

$$a_{27} = -5(27) + 3$$

$$= -132$$

16. -1809

$$S_{27} = 27 \left( \frac{-2 - 132}{2} \right)$$

$$= -1809$$

17.  $\sum_{m=1}^{31} (7m - 3)$

$$a_1 = 4; a_{31} = 214$$

$$S_{31} = 31 \left( \frac{4 + 214}{2} \right)$$

$$= 3379$$

17. 3379

18.  $\sum_{k=5}^{23} (1 - 2k)$

$$a_1 = -9, a_{19} = -45$$

$$S_{19} = 19 \left( \frac{-9 - 45}{2} \right)$$

$$= -513$$

18. -513

|        |        |
|--------|--------|
| Name:  | Date:  |
| Topic: | Class: |

| Main Ideas/Questions                                          | Notes/Examples                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GEOMETRIC</b><br><i>Sequence</i><br>Common<br><b>RATIO</b> | A sequence in which the ratio between any two terms is constant.                                                                                                                                                                                   |                                                                                                                                                                                                                                                                        |
|                                                               | The numerical ratio between any two terms. (Variable: $r$ )                                                                                                                                                                                        |                                                                                                                                                                                                                                                                        |
| <b>EXAMPLES</b>                                               | Determine whether the sequence is geometric. If yes, identify the common ratio and give the next three terms.                                                                                                                                      |                                                                                                                                                                                                                                                                        |
|                                                               | 1. $\{6, 12, 24, 48, \dots\}$<br>yes; $r = 2$<br>96, 192, 384                                                                                                                                                                                      | 2. $\{810, 270, 90, 30, \dots\}$<br>yes; $r = \frac{1}{3}$<br>$10, \frac{10}{3}, \frac{10}{9}$                                                                                                                                                                         |
|                                                               | 3. $\{-2, 10, -50, 250, \dots\}$<br>yes; $r = -5$<br>-1250, 6250, -31250                                                                                                                                                                           | 4. $\{4, 8, 20, 60, \dots\}$<br>No                                                                                                                                                                                                                                     |
|                                                               | 5. $\{4, -1, \frac{1}{4}, -\frac{1}{16}, \dots\}$<br>yes; $r = -\frac{1}{4}$<br>$\frac{1}{64}, -\frac{1}{256}, \frac{1}{1024}$                                                                                                                     | 6. $\{2, 4, 16, 256, \dots\}$<br>No                                                                                                                                                                                                                                    |
|                                                               | 7. $\{-2, -14, -98, -686, \dots\}$<br>yes; $r = 7$<br>-4802, -33614, -235298                                                                                                                                                                       | 8. $\{3.2, 8, 20, 50, \dots\}$<br>yes; $r = 2.5$<br>125, 312.5, 781.25                                                                                                                                                                                                 |
|                                                               | The $n^{\text{th}}$ term of any geometric sequence can be found using the formula:                                                                                                                                                                 |                                                                                                                                                                                                                                                                        |
| <b>GEOMETRIC</b><br><b>SEQUENCE</b><br><i>Formula</i>         | <div style="border: 1px solid black; padding: 10px; display: inline-block;"> <math display="block">a_n = a_1 \cdot r^{n-1}</math> </div> <p>where <math>a_1</math> is the <u>first term</u> and<br/> <math>r</math> is the <u>common ratio</u></p> |                                                                                                                                                                                                                                                                        |
| <b>EXAMPLES</b>                                               | Write a rule for each sequence, then find $a_6$ .                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                        |
|                                                               | 9. $\{1, 5, 25, 125, \dots\}$<br><div style="border: 1px solid black; padding: 5px; display: inline-block;"> <math>a_n = 1 \cdot (5)^{n-1}</math> </div><br>$a_6 = 1 \cdot (5)^{6-1}$<br>$= 1 \cdot 3125 = \boxed{3125}$                           | 10. $\{130, 65, 32.5, 16.25, \dots\}$<br><div style="border: 1px solid black; padding: 5px; display: inline-block;"> <math>a_n = 130 \cdot (\frac{1}{2})^{n-1}</math> </div><br>$a_6 = 130 \cdot (\frac{1}{2})^{6-1}$<br>$= 130 \cdot (\frac{1}{32}) = \boxed{4.0625}$ |

$$11. \{2, -8, 32, -128, \dots\}$$

$$a_n = 2 \cdot (-4)^{n-1}$$

$$\begin{aligned} a_6 &= 2 \cdot (-4)^{6-1} \\ &= 2 \cdot (-1024) \\ &= -2048 \end{aligned}$$

$$12. \{8, 24, 72, 216, \dots\}$$

$$a_n = 8 \cdot (3)^{n-1}$$

$$\begin{aligned} a_6 &= 8 \cdot (3)^{6-1} \\ &= 8 \cdot 243 \\ &= 1944 \end{aligned}$$

$$13. \{135, 90, 60, 40, \dots\}$$

$$a_n = 135 \cdot \left(\frac{2}{3}\right)^{n-1}$$

$$\begin{aligned} a_6 &= 135 \cdot \left(\frac{2}{3}\right)^{6-1} \\ &= 135 \cdot \frac{32}{243} \\ &= 17.\bar{7} \end{aligned}$$

$$14. \{-3200, 800, -200, 50, \dots\}$$

$$a_n = -3200 \cdot \left(-\frac{1}{4}\right)^{n-1}$$

$$\begin{aligned} a_6 &= -3200 \cdot \left(-\frac{1}{4}\right)^{6-1} \\ &= -3200 \cdot \left(-\frac{1}{1024}\right) \\ &= 3.125 \end{aligned}$$

Use the information given to find the indicated value.

$$15. a_1 = 7 \text{ and } r = 4; \text{ find } a_8$$

$$\begin{aligned} a_8 &= 7 \cdot (4)^{8-1} \\ &= 7 \cdot 16384 \\ &= 114,688 \end{aligned}$$

$$16. a_5 = -68 \text{ and } r = -0.5; \text{ find } a_1$$

$$\begin{aligned} -68 &= a_1 \cdot (-0.5)^{5-1} \\ -68 &= a_1 \cdot (0.0625) \\ -1088 &= a_1 \end{aligned}$$

$$17. a_1 = 13 \text{ and } a_4 = 4459; \text{ find } a_3$$

$$\begin{aligned} 4459 &= 13 \cdot (r)^{4-1} \\ 343 &= r^3 \\ 7 &= r \\ a_3 &= 13 \cdot (7)^{3-1} \\ &= 13 \cdot 49 \\ &= 637 \end{aligned}$$

$$18. a_2 = 6 \text{ and } a_5 = 0.048; \text{ find } a_4$$

$$\begin{aligned} a_5 &= a_2 \cdot r^{5-2} \\ 0.048 &= 6 \cdot r^3 \\ 0.008 &= r^3 \\ 0.2 &= r \\ a_4 &= 30 \cdot (0.2)^{4-1} \\ &= 30(0.008) \\ &= 0.24 \end{aligned}$$

## APPLICATION

Last year, Jackson bought a brand new car for \$47,500. If the car depreciates in value by 20% each year, what will the car be worth when it is 8 years old?

$$\begin{aligned} a_8 &= 38000 (0.8)^{8-1} \\ &= 38000 (0.2097152) \\ &= \$7969.18 \end{aligned}$$

Name: \_\_\_\_\_

Unit 10: Sequences &amp; Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 3: Geometric Sequences

\*\* This is a 2-page document! \*\*

Determine whether the sequence is geometric. If yes, identify the common ratio and give the next three terms.

1.  $\{2, 16, 128, \dots\}$

yes;  $r=8$ 

1024, 8192, 65536

2.  $\{2700, -900, 300, \dots\}$

yes;  $r = -\frac{1}{3}$ -100,  $33.\bar{3}$ , -11.1

3.  $\{5, 10, 30, \dots\}$

No

4.  $\{-3, 12, -48, \dots\}$

yes;  $r = -4$ 

192, -768, 3072

5.  $\{-1, -\frac{1}{2}, -\frac{1}{4}, \dots\}$

yes;  $r = \frac{1}{2}$  $-\frac{1}{8}, -\frac{1}{16}, -\frac{1}{32}$ 

6.  $\{8, 9, 10, \dots\}$

No

Write a rule for each sequence, then find  $a_8$ .

7.  $\{9, -18, 36, -72, \dots\}$

$$a_n = 9 \cdot (-2)^{n-1}$$

$$a_8 = 9 \cdot (-2)^{8-1}$$

$$= 9(-128)$$

$$= -1152$$

8.  $\{3, 15, 75, 375, \dots\}$

$$a_n = 3 \cdot (5)^{n-1}$$

$$a_8 = 3 \cdot (5)^{8-1}$$

$$= 3 \cdot 78125$$

$$= 234,375$$

9.  $\{6400, 1600, 400, 100, \dots\}$

$$a_n = 6400 \cdot \left(\frac{1}{4}\right)^{n-1}$$

$$a_8 = 6400 \cdot \left(\frac{1}{4}\right)^{8-1}$$

$$= 6400 \left(\frac{1}{16384}\right)$$

$$= 0.390625$$

10.  $\{-1.5, 4.5, -13.5, 40.5, \dots\}$

$$a_n = -1.5 \cdot (-3)^{n-1}$$

$$a_8 = -1.5(-3)^{8-1}$$

$$= -1.5(-2187)$$

$$= 3280.5$$

11.  $\{-2, -14, -98, -686, \dots\}$

$$a_n = -2 \cdot (7)^{n-1}$$

$$a_8 = -2 \cdot (7)^{8-1}$$

$$= -2(823543)$$

$$= -1,647,086$$

12.  $\left\{\frac{16}{81}, \frac{8}{27}, \frac{4}{9}, \frac{2}{3}, \dots\right\}$

$$a_n = \frac{16}{81} \cdot \left(\frac{3}{2}\right)^{n-1}$$

$$a_8 = \frac{16}{81} \cdot \left(\frac{3}{2}\right)^{8-1}$$

$$= \frac{16}{81} \cdot \left(\frac{3}{2}\right)^7 = 3.375$$

Use the information given to write a rule for the sequence and find the indicated value.

13.  $a_1 = 4$  and  $r = 6$ ; find  $a_7$

$$\begin{aligned} a_7 &= 4 \cdot (6)^{7-1} \\ &= 4 \cdot 46656 \\ &= 186,624 \end{aligned}$$

14.  $a_8 = 1024$  and  $r = -2$ ; find  $a_4$

$$\begin{aligned} 1024 &= a_1 \cdot (-2)^{8-1} \\ 1024 &= a_1 \cdot (-128) \\ -8 &= a_1 \\ a_4 &= -8 \cdot (-2)^{4-1} \\ &= -8 \cdot (-8) = 64 \end{aligned}$$

15.  $a_3 = -27$  and  $r = \frac{1}{3}$ ; find  $a_1$

$$\begin{aligned} -27 &= a_1 \cdot \left(\frac{1}{3}\right)^{3-1} \\ -27 &= a_1 \cdot \left(\frac{1}{9}\right) \\ -243 &= a_1 \end{aligned}$$

16.  $a_1 = 2400$  and  $a_4 = 37.5$ ; find  $a_2$

$$\begin{aligned} 37.5 &= 2400 \cdot r^{4-1} \\ 37.5 &= 2400 \cdot r^3 \\ 0.015625 &= r^3 \\ r &= 0.25 \\ a_2 &= 2400 \cdot (0.25)^{2-1} \\ &= 2400 \cdot 0.25 = 600 \end{aligned}$$

17.  $a_2 = 48$  and  $a_4 = 75$ ; find  $a_1$

$$\begin{aligned} a_4 &= a_2 \cdot r^{4-2} \\ 75 &= 48 \cdot r^2 \\ \frac{25}{16} &= r^2 \rightarrow r = \pm \frac{5}{4} \\ 48 &= a_1 \cdot \left(\pm \frac{5}{4}\right)^{2-1} \\ a_1 &= \pm 38.4 \end{aligned}$$

18.  $a_3 = 16$  and  $a_6 = 250$ ; find  $a_8$

$$\begin{aligned} a_6 &= a_3 \cdot r^{6-3} \\ 250 &= 16 \cdot r^3 \\ \frac{125}{8} &= r^3 \rightarrow r = \frac{5}{2} \\ a_8 &= 16 \cdot \left(\frac{5}{2}\right)^{8-3} \\ &= 1562.5 \end{aligned}$$

19. Jameson opened a new savings account with \$1500. The balance of the account for the first three months is given in the table below. Write a rule to represent the balance of her account, then estimate the balance of the account in 2 years.  $m = 24$

| Month | Balance (\$) |
|-------|--------------|
| 1     | 1500         |
| 2     | 1545         |
| 3     | 1591.35      |

$$r = 1.03$$

$$\begin{aligned} a_n &= 1500 \cdot (1.03)^{n-1} \\ a_{24} &= 1500 \cdot (1.03)^{24-1} \\ &= \$2960.38 \end{aligned}$$

20. Brady dropped a bouncy ball from a height of 64 feet. After the second bounce, the ball reached a height of 36 feet. Write a rule to represent the height of the ball after each bounce, then find the height after the fourth bounce.

$$a_0 = 64$$

$$a_2 = 36$$

$$a_2 = a_0 \cdot r^2$$

$$36 = 64 \cdot r^2$$

$$\frac{36}{64} = r^2$$

$$\frac{3}{4} = r$$

$$\begin{aligned} a_4 &= 48 \cdot \left(\frac{3}{4}\right)^3 \\ &= 48 \left(\frac{27}{64}\right) \\ &= 20.25 \text{ ft} \end{aligned}$$

|        |        |
|--------|--------|
| Name:  | Date:  |
| Topic: | Class: |

| Main Ideas/Questions              | Notes/Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GEOMETRIC</b><br><i>Series</i> | <p>A <b>geometric series</b> is the <u>Sum</u> of a geometric sequence.<br/>           To find the sum, use the following formula:</p> <div style="border: 1px solid black; padding: 10px; width: fit-content; margin: 10px auto;"> <math display="block">S_n = \frac{a_1 \cdot (1 - r^n)}{1 - r}</math> </div> <p>where <math>n</math> is the <u>number of terms</u>,<br/> <math>a_1</math> is the <u>first term</u>, and <math>r</math> is the <u>common ratio</u>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>EXAMPLES</b>                   | <p>Find the indicated sum for each geometric series.</p> <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;"> <p>1. <math>\{4 + 20 + 100 + 500 + \dots\}; S_{14}</math> <span style="float: right;"><math>n=14</math><br/><math>a_1=4</math><br/><math>r=5</math></span></p> <math display="block">S_{14} = \frac{4 \cdot (1 - 5^{14})}{1 - 5}</math> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;"> <math>= 6,103,515,624</math> </div> </div> <div style="width: 50%;"> <p>2. <math>\{81 + 27 + 9 + 3 + \dots\}; S_{20}</math> <span style="float: right;"><math>n=20</math><br/><math>a_1=81</math><br/><math>r=1/3</math></span></p> <math display="block">S_{20} = \frac{81 (1 - (1/3)^{20})}{1 - (1/3)}</math> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;"> <math>= 121.5</math> </div> </div> <div style="width: 50%;"> <p>3. <math>\{3 + 6 + 12 + 24 + \dots + 6144\}</math></p> <math display="block">6144 = 3 \cdot (2)^{n-1}</math> <math display="block">2048 = 2^{n-1}</math> <math display="block">11 = n - 1 \rightarrow 12 = n</math> <math display="block">S_{12} = \frac{3 (1 - 2^{12})}{1 - 2} = 12,285</math> </div> <div style="width: 50%;"> <p>4. <math>\{4 - 12 + 36 - 108 + \dots + 2916\}</math></p> <math display="block">2916 = 4 \cdot (-3)^{n-1}</math> <math display="block">729 = (-3)^{n-1}</math> <math display="block">6 = n - 1 \rightarrow 7 = n</math> <math display="block">S_7 = \frac{4 (1 - (-3)^7)}{1 - (-3)} = 2188</math> </div> <div style="width: 50%;"> <p>5. <math>\sum_{n=1}^{25} 2^{n-1}</math> <span style="float: right;"><math>r=2</math><br/><math>a_1=1</math><br/><math>n=25</math></span></p> <math display="block">S_{25} = \frac{1 (1 - 2^{25})}{1 - 2}</math> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;"> <math>= 33,554,431</math> </div> </div> <div style="width: 50%;"> <p>6. <math>\sum_{x=1}^8 -32 \cdot \left(-\frac{1}{2}\right)^{x-1}</math> <span style="float: right;"><math>r=-1/2</math><br/><math>n=8</math><br/><math>a_1=-32</math></span></p> <math display="block">S_8 = \frac{-32 (1 - (-1/2)^8)}{1 - (-1/2)}</math> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;"> <math>= -21.25</math> </div> </div> <div style="width: 50%;"> <p>7. <math>\sum_{k=6}^{20} -4 \cdot (-3)^{k-1}</math> <span style="float: right;"><math>r=-3</math><br/><math>n=15</math><br/><math>a_1=972</math></span></p> <math display="block">S_{15} = \frac{972 (1 - (-3)^{15})}{1 - (-3)}</math> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;"> <math>= 3,486,784,644</math> </div> </div> <div style="width: 50%;"> <p>8. <math>\sum_{m=2}^{11} 81 \cdot \left(\frac{1}{2}\right)^{m-1}</math> <span style="float: right;"><math>n=10</math><br/><math>r=1/2</math><br/><math>a_1=40.5</math></span></p> <math display="block">S_{10} = \frac{40.5 (1 - (1/2)^{10})}{1 - (1/2)}</math> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;"> <math>= \frac{82863}{1024} \approx 80.9029</math> </div> </div> </div> |

|                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|-----|-------|------|-------|-------|-------|--------|-------|---------|-------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|--|-------|-----|-------|-----|-------|-----|-------|-----|-------|------|-------|------|
| INFINITE<br>Geometric Series                                                                                                                                                               | Find the partial sums for each infinite series below:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
|                                                                                                                                                                                            | <table><tr><td colspan="2"><math>\left\{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots\right\}</math></td></tr><tr><td><math>S_1</math></td><td>0.5</td></tr><tr><td><math>S_2</math></td><td>0.75</td></tr><tr><td><math>S_3</math></td><td>0.875</td></tr><tr><td><math>S_4</math></td><td>0.9375</td></tr><tr><td><math>S_5</math></td><td>0.96875</td></tr><tr><td><math>S_6</math></td><td>0.984375</td></tr></table> $S_n \rightarrow 1$ <p>A series that approaches a certain sum is called a <b>CONVERGENT SERIES</b>.</p> | $\left\{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots\right\}$                                                                                                               |  | $S_1$ | 0.5 | $S_2$ | 0.75 | $S_3$ | 0.875 | $S_4$ | 0.9375 | $S_5$ | 0.96875 | $S_6$ | 0.984375 | <table><tr><td colspan="2"><math>\left\{\frac{1}{2} + 1 + 2 + 4 + 8 + \dots\right\}</math></td></tr><tr><td><math>S_1</math></td><td>0.5</td></tr><tr><td><math>S_2</math></td><td>1.5</td></tr><tr><td><math>S_3</math></td><td>3.5</td></tr><tr><td><math>S_4</math></td><td>7.5</td></tr><tr><td><math>S_5</math></td><td>15.5</td></tr><tr><td><math>S_6</math></td><td>31.5</td></tr></table> $S_n \rightarrow \infty$ <p>A series that does not have a certain sum is called a <b>DIVERGENT SERIES</b>.</p> | $\left\{\frac{1}{2} + 1 + 2 + 4 + 8 + \dots\right\}$ |  | $S_1$ | 0.5 | $S_2$ | 1.5 | $S_3$ | 3.5 | $S_4$ | 7.5 | $S_5$ | 15.5 | $S_6$ | 31.5 |
|                                                                                                                                                                                            | $\left\{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots\right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
|                                                                                                                                                                                            | $S_1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.5                                                                                                                                                                                           |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_2$                                                                                                                                                                                      | 0.75                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_3$                                                                                                                                                                                      | 0.875                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_4$                                                                                                                                                                                      | 0.9375                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_5$                                                                                                                                                                                      | 0.96875                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_6$                                                                                                                                                                                      | 0.984375                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $\left\{\frac{1}{2} + 1 + 2 + 4 + 8 + \dots\right\}$                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_1$                                                                                                                                                                                      | 0.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_2$                                                                                                                                                                                      | 1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_3$                                                                                                                                                                                      | 3.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_4$                                                                                                                                                                                      | 7.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_5$                                                                                                                                                                                      | 15.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| $S_6$                                                                                                                                                                                      | 31.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| <ul style="list-style-type: none"><li>If <math> r  &lt; 1</math>, then the series is <u>Convergent</u>.</li><li>If <math> r  &gt; 1</math>, then the series is <u>divergent</u>.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| Convergent Series<br><b>FORMULA</b>                                                                                                                                                        | To find the sum of a <b>convergent infinite geometric series</b> , use the formula: <div><math display="block">S_n = a_1 \left( \frac{1}{1-r} \right)</math></div>                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
| EXAMPLES                                                                                                                                                                                   | Determine if the series is convergent or divergent. If convergent, find the sum.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
|                                                                                                                                                                                            | 9. $\left\{1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots\right\}$ $r = \frac{1}{4}$ ;<br>Convergent!<br>$S_n = 1 \left( \frac{1}{1 - \frac{1}{4}} \right) = \boxed{\frac{4}{3}}$                                                                                                                                                                                                                                                                                                                                                           | 10. $\left\{\frac{2}{3} + \frac{8}{3} + \frac{32}{3} + \frac{128}{3} + \dots\right\}$<br>$r = 4$ ;<br>divergent                                                                               |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
|                                                                                                                                                                                            | 11. $\{5.8 + 3.48 + 2.088 + 1.2528 + \dots\}$<br>$r = 0.6$ ;<br>Convergent!<br>$S_n = 5.8 \left( \frac{1}{1 - 0.6} \right) = \boxed{14.5}$                                                                                                                                                                                                                                                                                                                                                                                                         | 12. $\{128 - 96 + 72 - 54 + \dots\}$<br>$r = -0.75$ ;<br>Convergent!<br>$S_n = 128 \left( \frac{1}{1 + .75} \right) = \boxed{\frac{512}{7}}$                                                  |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
|                                                                                                                                                                                            | 13. $\sum_{x=1}^{\infty} -2 \cdot \left(\frac{1}{4}\right)^{x-1}$ $r = \frac{1}{4}$ ;<br>Convergent!<br>$S_n = -2 \left( \frac{1}{1 - \frac{1}{4}} \right) = \boxed{-\frac{8}{3}}$                                                                                                                                                                                                                                                                                                                                                                 | 14. $\sum_{p=1}^{\infty} 7 \cdot (0.8)^{p-1}$ $r = 0.8$ ;<br>Convergent!<br>$S_n = 7 \left( \frac{1}{1 - .8} \right) = \boxed{35}$                                                            |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |
|                                                                                                                                                                                            | 15. $\sum_{w=4}^{\infty} 8 \cdot \left(-\frac{5}{2}\right)^{w-1}$<br>$r = -\frac{5}{2}$ ;<br>divergent!                                                                                                                                                                                                                                                                                                                                                                                                                                            | 16. $\sum_{k=2}^{\infty} -6 \cdot \left(-\frac{3}{4}\right)^{k-1}$ $r = -\frac{3}{4}$ ;<br>Convergent!<br>$S_n = \frac{9}{4} \left( \frac{1}{1 + \frac{3}{4}} \right) = \boxed{\frac{18}{7}}$ |  |       |     |       |      |       |       |       |        |       |         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |  |       |     |       |     |       |     |       |     |       |      |       |      |



Name: \_\_\_\_\_

Unit 10: Sequences &amp; Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 4: Geometric Series

\*\* This is a 2-page document! \*\*

$$S_n = \frac{a_1(1-r^n)}{1-r}$$

Find the indicated sum for each geometric series.

1.  $\{2+6+18+54+\dots\}; S_{12}$   $r=3$

$$S_{12} = \frac{2(1-3^{12})}{1-3}$$

$$= \boxed{531,440}$$

2.  $\{18-9+\frac{9}{2}-\frac{9}{4}+\dots\}; S_8$   $r=-\frac{1}{2}$

$$S_8 = \frac{18(1-(-\frac{1}{2})^8)}{1-(-\frac{1}{2})}$$

$$= \boxed{\frac{765}{64}}$$

3.  $\{10+1+\frac{1}{10}+\frac{1}{100}+\dots\}; S_7$   $r=\frac{1}{10}$

$$S_n = \frac{10(1-(\frac{1}{10})^7)}{1-\frac{1}{10}}$$

$$= \boxed{11.\bar{1}}$$

4.  $\{-\frac{1}{8}+\frac{1}{2}-2+8+\dots\}; S_{10}$   $r=-4$

$$S_{10} = \frac{-\frac{1}{8}(1-(-4)^{10})}{1-(-4)}$$

$$= \boxed{26214.375}$$

5.  $\{(-1)+(-6)+(-36)+\dots+(-7776)\}$   $r=6$

$$-7776 = -1(6)^{n-1}$$

$$7776 = 6^{n-1}$$

$$5 = n-1 \rightarrow n=6$$

$$S_6 = \frac{-1(1-6^6)}{1-6} = \boxed{-9331}$$

6.  $\{1875+750+300+\dots+3.072\}$   $r=0.4$

$$3.072 = 1875(0.4)^{n-1}$$

$$0.0016384 = (0.4)^{n-1}$$

$$7 = n-1 \rightarrow n=8$$

$$S_8 = \frac{1875(1-0.4^8)}{1-0.4} = \boxed{3122.952}$$

7.  $\sum_{x=1}^{12} (-4)^{x-1}$   $r=-4$   
 $a_1=1$   
 $n=12$

$$S_{12} = \frac{1(1-(-4)^{12})}{1-(-4)}$$

$$= \boxed{-3,355,443}$$

8.  $\sum_{m=1}^7 24 \cdot \left(\frac{1}{2}\right)^{m-1}$   $r=\frac{1}{2}$   
 $a_1=24$   
 $n=7$

$$S_7 = \frac{24(1-(\frac{1}{2})^7)}{1-\frac{1}{2}}$$

$$= \boxed{47.625}$$

9.  $\sum_{k=4}^{14} 81 \cdot \left(-\frac{4}{3}\right)^{k-1}$   $r=-\frac{4}{3}$   
 $a_1=-192$   
 $n=11$

$$S_{11} = \frac{-192(1-(-\frac{4}{3})^{11})}{1-(-\frac{4}{3})}$$

$$= \boxed{-2630.562}$$

10.  $\sum_{w=2}^9 \frac{1}{10} \cdot (5)^{w-1}$   $r=5$   
 $a_1=0.5$   
 $n=8$

$$S_8 = \frac{0.5(1-5^8)}{1-5}$$

$$= \boxed{48,828}$$

Determine whether each geometric sequence is convergent or divergent.  
Find the sum when possible.

$$S_n = \frac{a_1}{1-r}$$

11.  $\left\{180 - 60 + 20 - \frac{20}{3} + \dots\right\}$   $r = -\frac{1}{3}$ ;  
Convergent

$$S_n = \frac{180}{1 - (-\frac{1}{3})} = \boxed{135}$$

12.  $\{250 + 200 + 160 + 128 + \dots\}$   $r = 0.8$ ;  
Convergent

$$S_n = \frac{250}{1 - 0.8} = \boxed{1250}$$

13.  $\left\{3 + \frac{9}{2} + \frac{27}{4} + \frac{81}{8} + \dots\right\}$   $r = \frac{3}{2}$ ;  
divergent

14.  $\left\{-\frac{8}{3} + \frac{4}{3} - \frac{2}{3} + \frac{1}{3} + \dots\right\}$   $r = -0.5$ ;  
Convergent

$$S_n = \frac{-\frac{8}{3}}{1 - (-0.5)} = \boxed{-1.\bar{7}}$$

15.  $\sum_{k=1}^{\infty} -\frac{1}{4} \cdot \left(-\frac{1}{5}\right)^{k-1}$   $r = -\frac{1}{5}$ ;  
Convergent

$$S_n = \frac{-\frac{1}{4}}{1 - (-\frac{1}{5})} = \boxed{-0.208\bar{3}}$$

16.  $\sum_{n=1}^{\infty} \frac{1}{2} \cdot 5^{n-1}$   $r = 5$ ;  
divergent

17.  $\sum_{y=1}^{\infty} -6.8 \cdot \left(\frac{3}{5}\right)^{y-1}$   $r = \frac{3}{5}$ ;  
Convergent

$$S_n = \frac{-6.8}{1 - \frac{3}{5}} = \boxed{-17}$$

18.  $\sum_{m=1}^{\infty} 18 \cdot (0.25)^{m-1}$   $r = 0.25$ ;  
Convergent

$$S_n = \frac{18}{1 - 0.25} = \boxed{24}$$

19.  $\sum_{p=1}^{\infty} 10 \cdot \left(-\frac{4}{3}\right)^{p-1}$   $r = -\frac{4}{3}$ ;  
divergent

20.  $\sum_{x=1}^{\infty} -128 \cdot \left(-\frac{1}{4}\right)^{x-1}$   $r = -\frac{1}{4}$ ;  
Convergent

$$S_n = \frac{-128}{1 - (-\frac{1}{4})} = \boxed{-102.4}$$

# ARITHMETIC, GEOMETRIC... or Neither?

**Directions:** Determine the type of sequence. Shade the boxes using the color codes below.  
For arithmetic and geometric sequences, identify the common difference or common ratio, then write the formula.

$$a_n = d(n-1) + a_1$$

$$a_n = a_1 \cdot r^{n-1}$$

**Arithmetic Sequence:** Purple

**Geometric Sequence:** Light Green

**Neither:** Uncolored

|                                                                                             |                                                                                                                                                            |                                                                                                                                                             |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. $\{4, -20, 100, -500, \dots\}$<br>Geo; $r = -5$<br>$a_n = 4 \cdot (-5)^{n-1}$            | 2. $\{15, 5, \frac{5}{3}, \frac{5}{9}, \dots\}$<br>Geo; $r = \frac{1}{3}$<br>$a_n = 15 \cdot (\frac{1}{3})^{n-1}$                                          | 3. $\{1, -2, 3, -4, \dots\}$<br>$\text{Neither}$                                                                                                            |
| 4. $\{-18, -11, -4, 3, \dots\}$<br>Arith; $d = 7$<br>$a_n = 7(n-1) - 18$<br>$a_n = 7n - 25$ | 5. $\{1, 8, 27, 64, \dots\}$<br>$\text{Neither}$                                                                                                           | 6. $\{\frac{9}{4}, \frac{5}{2}, \frac{11}{4}, 3, \dots\}$<br>Arith; $d = \frac{1}{4}$<br>$a_n = \frac{1}{4}(n-1) + \frac{9}{4}$<br>$a_n = \frac{1}{4}n + 2$ |
| 7. $\{-3, -27, -243, -2187, \dots\}$<br>Geo; $r = 9$<br>$a_n = -3 \cdot (9)^{n-1}$          | 8. $\{-5, -\frac{17}{2}, -12, -\frac{31}{2}, \dots\}$<br>Arith; $d = -\frac{7}{2}$<br>$a_n = -\frac{7}{2}(n-1) - 5$<br>$a_n = -\frac{7}{2}n - \frac{3}{2}$ | 9. $\{100, 40, 16, \frac{32}{5}, \dots\}$<br>Geo; $r = \frac{2}{5}$<br>$a_n = 100 \cdot (\frac{2}{5})^{n-1}$                                                |

Arith:  $S_n = n \left( \frac{a_1 + a_n}{2} \right)$  **SERIES REVIEW** Geo:  $S_n = \frac{a_1(1-r^n)}{1-r}$

|                                                                                                                                          |                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Directions:</b> Indicate whether the series is arithmetic or geometric, then find the sum.                                            |                                                                                                                                                                          |
| 10. $\{3, -1, -5, -9, \dots\}; S_{26}$ Arith<br>$a_{26} = -4(26-1) + 3$ $= -97$ $S_n = 26 \left( \frac{3-97}{2} \right) = \boxed{-1222}$ | 11. $\{-60, -30, -15, -\frac{15}{2}, \dots\}; S_{14}$ Geo<br>$S_n = \frac{-60(1-(\frac{1}{2})^{14})}{1-(\frac{1}{2})} = \boxed{\frac{-245745}{2048}}$ $\approx (119.99)$ |
| 12. $\sum_{k=1}^9 -3 \cdot 2^{k-1}$ Geo<br>$S_n = \frac{-3(1-2^9)}{1-2} = \boxed{-1533}$                                                 | 13. $\sum_{x=4}^{39} (5x+2)$ Arith<br>$a_1 = 22, a_{36} = 197, n = 36$ $S_n = 36 \left( \frac{22+197}{2} \right) = \boxed{3942}$                                         |

|                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>14. <math>\sum_{n=4}^{22} \left(1 - \frac{1}{3}n\right)</math> Arith</p> <p><math>a_1 = -\frac{1}{3}, a_9 = -\frac{19}{3}, n=19</math></p> <p><math>S_n = 19 \left(\frac{-\frac{1}{3} - \frac{19}{3}}{2}\right) = \boxed{-\frac{190}{3}}</math></p> | <p>15. <math>\sum_{n=2}^{10} -16 \cdot \left(\frac{1}{2}\right)^{n-1}</math> Geo</p> <p><math>a_1 = -8, n=9</math></p> <p><math>S_n = \frac{-8(1 - (\frac{1}{2})^9)}{1 - \frac{1}{2}} = \boxed{-\frac{511}{32}}</math></p> |
| <p>16. <math>\sum_{y=1}^{\infty} -5 \cdot \left(-\frac{3}{4}\right)^{y-1}</math> Geo;<br/><math>n = \infty</math></p> <p><math>r = -3/4</math>; Convergent</p> <p><math>S_n = \frac{-5}{1 - (-3/4)} = \boxed{-\frac{20}{7}}</math></p>                 | <p>17. <math>\sum_{n=4}^{\infty} \frac{1}{3} \cdot 6^{n-1}</math> Geo;<br/><math>n = \infty</math></p> <p><math>r = 6</math>; <span style="border: 1px solid black; padding: 2px;">divergent</span></p>                    |

## APPLICATIONS!

18. Ian is playing a video game. The number of points he has earned after every hour is shown below. Determine whether this represents an arithmetic or geometric sequence. Then find the approximate number of points earned after 9 hours. Round your answer to the nearest hundredth.

| Hour | Points |
|------|--------|
| 1    | 384    |
| 2    | 480    |
| 3    | 600    |
| 4    | 750    |

Geometric;  $r = \frac{5}{4}$

$$a_9 = 384 \cdot \left(\frac{5}{4}\right)^8$$

$$= \boxed{2288.82 \text{ points}}$$

19. Mikayla is building a pyramid of paper cups. The bottom row has 61 cups while the top row has only one cup. If there are 16 total rows and the number of cups on each row represents an arithmetic sequence, find the total number of cups.

Arithmetic

$$a_1 = 61$$

$$a_n = 1$$

$$n = 16$$

$$S_{16} = 16 \left(\frac{61 + 1}{2}\right)$$

$$= \boxed{496 \text{ cups}}$$

20. The amount of interest that Ben earned on his savings account for the past three years since opening the account is shown in the table below. Determine whether this pattern represents an arithmetic or geometric sequence. Then, find the total amount of interest that Ben will earn after 20 years.

| Year | Interest |
|------|----------|
| 1    | \$72     |
| 2    | \$79.20  |
| 3    | \$87.12  |

Geometric;  $r = 1.1$

$$S_n = \frac{72(1 - 1.1^{20})}{1 - 1.1} = \boxed{\$4123.80}$$

Name: \_\_\_\_\_

Unit 10: Sequences &amp; Series

Date: \_\_\_\_\_ Per: \_\_\_\_\_

Homework 5: Arithmetic & Geometric  
Sequences/Series (Mixed Review)**\*\* This is a 2-page document! \*\*****Determine whether the sequence is arithmetic, geometric, or neither. For arithmetic and geometric sequences, identify the common difference or common ratio, then write the formula.**

1.  $\{1, 2, 6, 24, 120, \dots\}$

Neither

2.  $\{31, 26, 21, 16, 11, \dots\}$

Arith;  $d = -5$ 

$$a_n = -5(n-1) + 31$$

$$a_n = -5n + 36$$

3.  $\{3888, -648, 108, -18, 3, \dots\}$

Geo;  $r = -\frac{1}{6}$ 

$$a_n = 3888 \cdot \left(-\frac{1}{6}\right)^{n-1}$$

4.  $\left\{-2, -\frac{7}{3}, -\frac{8}{3}, -3, \dots\right\}$

Arith;  $d = -\frac{1}{3}$ 

$$a_n = -\frac{1}{3}(n-1) - 2$$

$$a_n = -\frac{1}{3}n - \frac{5}{3}$$

5.  $\{-14, -28, -56, -112, \dots\}$

Geo;  $r = 2$ 

$$a_n = -14 \cdot (2)^{n-1}$$

6.  $\{0, -1, -3, -6, -10, \dots\}$

Neither

7.  $\{-45, -38, -31, -24, \dots\}$

Arith;  $d = 7$ 

$$a_n = 7(n-1) - 45$$

$$a_n = 7n - 52$$

8.  $\left\{\frac{5}{9}, \frac{5}{6}, \frac{5}{4}, \frac{15}{8}, \frac{45}{16}, \dots\right\}$

Geo;  $r = \frac{3}{2}$ 

$$a_n = \frac{5}{9} \cdot \left(\frac{3}{2}\right)^{n-1}$$

9.  $\left\{\frac{1}{32}, -\frac{1}{8}, \frac{1}{2}, -2, 8, \dots\right\}$

Geo;  $r = -4$ 

$$a_n = \frac{1}{32} \cdot (-4)^{n-1}$$

**Determine whether the series is arithmetic or geometric, then find the indicated sum.**

10.  $\{-6561 + 2187 - 729 + 243 + \dots\}; S_9$  Geo;  $r = -\frac{1}{3}$

$$S_n = \frac{-6561(1 - (-\frac{1}{3})^9)}{1 - (-\frac{1}{3})} = -4921$$

11.  $\left\{-\frac{3}{2} + 1 + \frac{7}{2} + 6 + \frac{17}{2} + \dots\right\}; S_{24}$  Arith;  $d = \frac{5}{2}$

$$a_{24} = \frac{5}{2}(24-1) - \frac{3}{2} = 56$$

$$S_n = 24 \left( \frac{-\frac{3}{2} + 56}{2} \right) = 654$$

12.  $\sum_{p=1}^{34} (19 - 2p)$

Arith

$$S_n = 34 \left( \frac{17 - 49}{2} \right) = -544$$

13.  $\sum_{n=1}^7 -\frac{1}{8} \cdot 4^{n-1}$

Geo;  $r = 4$ 

$$S_n = \frac{-\frac{1}{8}(1 - 4^7)}{1 - 4} = -\frac{5461}{8}$$

$$14. \sum_{x=2}^{10} 640 \cdot \left(\frac{1}{2}\right)^{x-1} \quad \text{Geo; } a_1 = 320, n = 9$$

$$S_n = \frac{320(1 - (\frac{1}{2})^9)}{1 - \frac{1}{2}} = \boxed{638.75}$$

$$15. \sum_{k=1}^{29} (5k - 42) \quad \text{Arith}$$

$$S_n = 29 \left( \frac{-37 + 103}{2} \right) = \boxed{957}$$

$$16. \sum_{m=1}^{\infty} 250 \cdot \left(\frac{1}{5}\right)^{m-1} \quad \text{Geo; } r = \frac{1}{5} \text{ conv.}$$

$$S_n = \frac{250}{1 - \frac{1}{5}} = \boxed{312.5}$$

$$17. \sum_{y=1}^{\infty} -2 \cdot \left(-\frac{2}{3}\right)^{y-1} \quad \text{Geo; } r = -\frac{2}{3} \text{ conv.}$$

$$S_n = \frac{-2}{1 - (-\frac{2}{3})} = \boxed{-\frac{6}{5}}$$

Solve each problem.

18. The football team is running a jackpot in which the prize will start at \$2500 for the first game, and then increase \$500 each game if there is no winner. If there are no winners, find the value of the jackpot at the end of their 16-game season.

Arithmetic

$$a_1 = 2500$$

$$d = 500$$

$$n = 16$$

$$a_n = 500(16-1) + 2500 = \boxed{\$10,000}$$

19. The value of Casey's home for the first three years after purchasing is given in the table below. If this pattern continues, find the approximate value of his home after 30 years.

| Year | Value     |
|------|-----------|
| 1    | \$185,400 |
| 2    | \$190,962 |
| 3    | \$196,691 |

Geometric  
 $r = 1.03$

$$a_n = 185400(1.03)^{29} = \boxed{\$436,907.24}$$

20. The construction crew is pounding a stake into the ground. On the first strike, it goes 24 centimeters into the ground, 18 centimeters further on the second strike, then 13.5 centimeters further on the third strike. If this pattern continues, find the total distance the stake was driven after seven strikes.

Geometric  
 $r = 0.75$

$$S_7 = \frac{24(1 - 0.75^7)}{1 - 0.75} = \boxed{83.19 \text{ cm}}$$

21. The table below shows the amount of snowfall for the first three days of January in the town of Orchard Hills. If this pattern continues, find the total amount of snowfall for the month of January.

| Day | Inches |
|-----|--------|
| 1   | 0.2    |
| 2   | 0.5    |
| 3   | 0.8    |

Arithmetic  
 $d = 0.3$   
 $n = 31$   
 $a_1 = 0.2$

$$a_{31} = 0.3(31-1) + 0.2 = 9.2$$

$$S_n = 31 \left( \frac{0.2 + 9.2}{2} \right) = \boxed{145.7 \text{ in}}$$

|            | ARITHMETIC<br>Sequence                                                                                                                                                                  | GEOMETRIC<br>Sequence                                                                                                                                                         | ARITHMETIC<br>Series                                                                                                                                                                              | GEOMETRIC<br>Series                                                                                                                                                                                 | INF. GEOMETRIC<br>Series                                                                                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition | A list of numbers with a common difference.                                                                                                                                             | A list of numbers with a common ratio.                                                                                                                                        | The sum of the numbers in an arithmetic sequence.                                                                                                                                                 | The sum of the finite list of numbers in a geometric sequence.                                                                                                                                      | The sum of elements in a converging ( $ r  < 1$ ) geometric sequence.                                                                                                                        |
| Formula    | $a_n = d(n-1) + a_1$                                                                                                                                                                    | $a_n = a_1 \cdot r^{n-1}$                                                                                                                                                     | $S_n = n \left( \frac{a_1 + a_n}{2} \right)$                                                                                                                                                      | $S_n = \frac{a_1(1-r^n)}{1-r}$                                                                                                                                                                      | $S_n = \frac{a_1}{1-r}$                                                                                                                                                                      |
| Example 1  | Find $a_{28}$ :<br>$\{-5, 9, 23, 37, 51, \dots\}$<br>$a_1 = -5, n = 28, d = 14$<br>$a_{28} = 14(28-1) - 5$<br>$\boxed{= 373}$                                                           | Find $a_{14}$ :<br>$\{-8, 16, -32, 64, \dots\}$<br>$a_1 = -8, r = -2$<br>$a_{14} = -8(-2)^{14-1}$<br>$\boxed{= 65,536}$                                                       | Find $S_{37}$ :<br>$\{(-3) + (-7) + (-11) + \dots\}$<br>$a_1 = -3, n = 37, d = -4$<br>$a_{37} = -4(37-1) - 3$<br>$= -147$<br>$S_{37} = 37 \left( \frac{-3 - 147}{2} \right)$<br>$\boxed{= -2775}$ | Find $S_{12}$ :<br>$\{2 - 6 + 18 - 54 + \dots\}$<br>$a_1 = 2, r = -3, n = 12$<br>$S_{12} = \frac{2(1-(-3)^{12})}{1-(-3)}$<br>$\boxed{= -265,720}$                                                   | Find the sum:<br>$\{400 + 200 + 100 + \dots\}$<br>$r = \frac{1}{2}, a_1 = 400$<br>$S_n = \frac{400}{1-\frac{1}{2}}$<br>$\boxed{= 800}$                                                       |
| Example 2  | Find $a_{23}$ :<br>$\left\{9, \frac{13}{2}, 4, \frac{3}{2}, -1, \dots\right\}$<br>$a_1 = 9, n = 23, d = -\frac{5}{2}$<br>$a_{23} = -\frac{5}{2}(23-1) - \frac{5}{2}$<br>$\boxed{= -46}$ | Find $a_7$ :<br>$\{1024, 768, 576, 432, \dots\}$<br>$a_1 = 1024, r = \frac{3}{4}$<br>$a_7 = 1024 \left( \frac{3}{4} \right)^{7-1}$<br>$\boxed{= \frac{729}{4}}$<br>$(182.25)$ | Find the sum:<br>$\sum_{k=1}^{33} (6k-15)$<br>$a_1 = -9, a_{33} = 183$<br>$n = 33$<br>$S_{33} = 33 \left( \frac{-9 + 183}{2} \right)$<br>$\boxed{= 2871}$                                         | Find the sum:<br>$\sum_{m=2}^8 -512 \cdot \left( \frac{3}{2} \right)^{m-1}$<br>$a_1 = -768, r = \frac{3}{2}, n = 7$<br>$S_n = \frac{-768(1-(\frac{3}{2})^7)}{1-\frac{3}{2}}$<br>$\boxed{= -24,708}$ | Find the sum:<br>$\sum_{y=1}^{\infty} 48 \cdot \left( -\frac{1}{4} \right)^{y-1}$<br>$r = -\frac{1}{4}, a_1 = 48$<br>$S_n = \frac{48}{1-(-\frac{1}{4})}$<br>$\boxed{= \frac{192}{5}}$ (38.4) |

# Unit 10 Test Study Guide

## (Sequences & Series)

Name: \_\_\_\_\_

Date: \_\_\_\_\_ Per: \_\_\_\_\_

### Topic 1: Sequences, Explicit & Recursive Formulas

|                                                                                                                                                                                               |                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1. What is a sequence? <u>A list of numbers that follow a particular pattern.</u>                                                                                                             |                                                                                                           |
| 2. Describe the difference between recursive and explicit formulas: <u>A recursive formula requires knowledge of previous terms, where an explicit formula only requires the term number.</u> |                                                                                                           |
| Given the formula, write the first six terms of each sequence.                                                                                                                                |                                                                                                           |
| 3. $a_1 = -1$ ; $a_n = 2a_{n-1} - 5$ (for $n \geq 2$ )<br><u>-1, -7, -19, -43, -91, -187</u>                                                                                                  | 4. $a_1 = 1$ , $a_2 = 2$ ; $a_n = (a_{n-1})^2 - a_{n-2}$ (for $n \geq 3$ )<br><u>1, 2, 3, 7, 46, 2109</u> |
| 5. $a_n = 3n - 8$<br><u>-5, -2, 1, 4, 7, 10</u>                                                                                                                                               | 6. $a_n = \frac{1}{4} \cdot (-2)^{n-1}$<br><u><math>\frac{1}{4}, -\frac{1}{2}, 1, -2, 4, -8</math></u>    |
| 7. $a_n = \frac{n+4}{n+2}$ <u><math>\frac{5}{3}, \frac{3}{2}, \frac{7}{5}, \frac{4}{3}, \frac{9}{7}, \frac{5}{4}</math></u>                                                                   | 8. $a_n = n^3 - n^2$<br><u>0, 4, 18, 48, 100, 180</u>                                                     |

### Topic 2: Expanding & Evaluating Series

|                                                                                                                                                                                        |                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 9. What is a series? <u>The sum of the terms in a sequence.</u>                                                                                                                        |                                                                                                              |
| Write each sequence as a series, then find $S_8$ .                                                                                                                                     |                                                                                                              |
| 10. $a_1 = 2$ , $a_2 = 3$ ; $a_n = a_{n-1} \cdot a_{n-2}$ (for $n \geq 3$ )<br><u><math>2 + 3 + 6 + 18 + 108 + 1944 + 209952 + 408146688</math></u><br><u><math>= 408358721</math></u> | 11. $a_n = n^2 + 2n$<br><u><math>3 + 8 + 15 + 24 + 35 + 48 + 63 + 80</math></u><br><u><math>= 276</math></u> |
| Expand and evaluate each series.                                                                                                                                                       |                                                                                                              |
| 12. $\sum_{k=1}^{16} (k-5)^2 = 16 + 9 + 4 + 1 + 0 + 1 + 4 + 8 + 16 + 25 + 36 + 49 + 64 + 81 + 100 + 121$<br><u><math>= 536</math></u>                                                  |                                                                                                              |
| 13. $\sum_{m=3}^{14} (m^3 - 4m) = 15 + 48 + 105 + 192 + 315 + 480 + 693 + 960 + 1287 + 1680 + 2145 + 2688$<br><u><math>= 10,608</math></u>                                             |                                                                                                              |



### Topic 3: Arithmetic vs. Geometric Sequences

|                                                                                                                                                              | DEFINITION                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| ARITHMETIC                                                                                                                                                   | A sequence with a common difference.                                                    |
| GEOMETRIC                                                                                                                                                    | A sequence with a common ratio.                                                         |
| For each sequence: a) Determine if the sequence is arithmetic, geometric, or neither and b) Determine the common difference ( $d$ ) or common ratio ( $r$ ). |                                                                                         |
| 14. $\{3, 12, 48, 192, 768, \dots\}$<br>Geometric; $r=4$                                                                                                     | 15. $\{16, 106, 1006, 10006, 100006, \dots\}$<br>Neither                                |
| 16. $\{2, -5, -12, -19, -26, \dots\}$<br>Arithmetic; $d = -7$                                                                                                | 17. $\{45, -15, 5, -\frac{5}{3}, \frac{5}{9}, \dots\}$<br>Geometric; $r = -\frac{1}{3}$ |
| 18. $\{6, 8, 11, 15, 20, \dots\}$<br>Neither                                                                                                                 | 19. $\{-11, -8, -5, -2, 1, \dots\}$<br>Arithmetic; $d=3$                                |
| 20. $\{-1125, -450, -180, -72, -\frac{144}{5}, \dots\}$<br>Geometric; $r = \frac{2}{5}$                                                                      | 21. $\{0, 1, 1, 2, 3, 5, \dots\}$<br>Neither                                            |

### Topic 4: Writing Arithmetic & Geometric Sequence Formulas

| Arithmetic Sequence Formula:<br>$a_n = d(n-1) + a_1$                                                                                                                                                                                            | Geometric Sequence Formula:<br>$a_n = a_1 \cdot r^{n-1}$                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Write a formula for each arithmetic sequence, then find the indicated term.                                                                                                                                                                     |                                                                                                                                                                                                     |
| 22. $\{-13, -7, -1, 5, \dots\}; a_{42}$<br>$a_n = 6(n-1) - 13$<br>$\quad = 6n - 19$<br>$a_{42} = 6(42) - 19 = \boxed{233}$                                                                                                                      | 23. $\{34, 24, 14, 4, \dots\}; a_{30}$<br>$a_n = -10(n-1) + 34$<br>$\quad = -10n + 44$<br>$a_{30} = -10(30) + 44 = \boxed{-256}$                                                                    |
| 24. $\{-\frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \dots\}; a_{21}$<br>$a_n = \frac{1}{4}(n-1) - \frac{3}{8}$<br>$\quad = \frac{1}{4}n - \frac{5}{8}$<br>$a_{21} = \frac{1}{4}(21) - \frac{5}{8} = \boxed{\frac{37}{8}}$ | 25. $\{4, \frac{5}{2}, 1, -\frac{1}{2}, -2, \dots\}; a_{25}$<br>$a_n = -\frac{3}{2}(n-1) + 4$<br>$\quad = -\frac{3}{2}n + \frac{11}{2}$<br>$a_{25} = -\frac{3}{2}(25) + \frac{11}{2} = \boxed{-32}$ |

Write a formula for the arithmetic sequence and find the indicated value.

26.  $a_{31} = -108$  and  $d = -3$ ; find  $a_1$

$$-108 = -3(31-1) + a_1$$

$$-108 = -90 + a_1$$

$$\boxed{-18 = a_1}$$

27.  $a_{14} = 15$  and  $d = 9$ ; find  $a_4$

$$15 = 9(14-1) + a_1$$

$$15 = 117 + a_1$$

$$\underline{\quad} \quad \underline{-102 = a_1} \quad \underline{\quad}$$

$$a_4 = 9(4-1) - 102$$

$$\boxed{= -75}$$

28.  $a_1 = -2$  and  $a_{10} = 43$ ; find  $d$

$$43 = d(10-1) - 2$$

$$45 = 9d$$

$$\boxed{5 = d}$$

29.  $a_1 = 10$  and  $a_{21} = 2$ ; find  $a_{14}$

$$2 = d(21-1) + 10$$

$$-8 = 20d$$

$$\underline{\quad} \quad \underline{d = -0.4} \quad \underline{\quad}$$

$$a_{14} = -0.4(14-1) + 10 = \boxed{4.8}$$

Write a formula for each geometric sequence, then find the indicated term.

30.  $\{8, -16, 32, -64, \dots\}; a_{18}$

$$\boxed{a_n = 8(-2)^{n-1}}$$

$$a_{18} = 8(-2)^{18-1}$$

$$\boxed{= -1,048,576}$$

31.  $\{6561, 2187, 729, 243, \dots\}; a_{11}$

$$\boxed{a_n = 6561 \cdot \left(\frac{1}{3}\right)^{n-1}}$$

$$a_{11} = 6561 \cdot \left(\frac{1}{3}\right)^{11-1}$$

$$\boxed{= \frac{1}{9}}$$

32.  $\left\{18, 24, 32, \frac{128}{3}, \dots\right\}; a_6$

$$\boxed{a_n = 18 \cdot \left(\frac{4}{3}\right)^{n-1}}$$

$$a_6 = 18 \cdot \left(\frac{4}{3}\right)^{6-1} = \boxed{\frac{2048}{27}}$$

33.  $\left\{-\frac{2}{5}, -2, -10, -50, \dots\right\}; a_8$

$$\boxed{a_n = -\frac{2}{5} \cdot (5)^{n-1}}$$

$$a_8 = -\frac{2}{5} \cdot (5)^{8-1} = \boxed{-31,250}$$

Write a formula for the geometric sequence and find the indicated value.

34.  $a_5 = -10$  and  $r = -\frac{1}{2}$ ; find  $a_1$

$$-10 = a_1 \cdot \left(-\frac{1}{2}\right)^{5-1}$$

$$-10 = 0.0625 \cdot a_1$$

$$\boxed{-160 = a_1}$$

35.  $a_1 = 2$  and  $a_5 = 4802$ ; find  $r$

$$4802 = 2 \cdot r^{5-1}$$

$$2401 = r^4$$

$$\boxed{7 = r}$$

36.  $a_3 = -18$  and  $a_6 = 486$ ; find  $a_1$

$$a_6 = a_3 \cdot r^{6-3}$$

$$486 = -18 \cdot r^3$$

$$-27 = r^3$$

$$-3 = r$$

$$a_3 = -18$$

$$a_2 = 6$$

$$\boxed{a_1 = -2}$$

37.  $a_2 = 1500$  and  $a_4 = 960$ ; find  $a_6$

$$960 = 1500 \cdot r^{4-2}$$

$$\frac{16}{25} = r^2$$

$$\frac{4}{5} = r$$

$$a_6 = 1500 \cdot \left(\frac{4}{5}\right)^{6-1}$$

$$\boxed{a_6 = 614.4}$$

**Topic 5: Arithmetic & Geometric Series**

| Sum of an Arithmetic Series<br>$S_n = n \left( \frac{a_1 + a_n}{2} \right)$                                                                                                                                        | Sum of a Geometric Series<br>$S_n = \frac{a_1 (1 - r^n)}{1 - r}$                                                                                                                                            | Sum of a Convergent Infinite Geometric Series<br>$ r  < 1; S_n = \frac{a_1}{1 - r}$ |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Find the indicated sum for each series.</b>                                                                                                                                                                     |                                                                                                                                                                                                             |                                                                                     |
| <b>38.</b> $\{6 + 1 + (-4) + (-9) + \dots\}; S_{22}$ Arith<br>$a_1 = 6, n = 22, d = -5$<br>$a_{22} = -5(21) + 6$<br>$= -99$<br>$S_{22} = 22 \left( \frac{6 - 99}{2} \right) = \boxed{-1023}$                       | <b>39.</b> $\{2 + 12 + 72 + 432 + \dots\}; S_{13}$ Geo<br>$a_1 = 2, r = 6, n = 13$<br>$S_{13} = \frac{2(1 - 6^{13})}{1 - 6}$<br>$= \boxed{5,224,277,606}$                                                   |                                                                                     |
| <b>40.</b> $\{800 - 200 + 50 - \frac{25}{2} + \dots\}; S_8$ Geo<br>$a_1 = 800, r = -\frac{1}{4}, n = 8$<br>$S_8 = \frac{800(1 - (-\frac{1}{4})^8)}{1 - (-\frac{1}{4})} = \boxed{\frac{327675}{512}}$<br>$(639.99)$ | <b>41.</b> $\{(-29) + (-27) + (-25) + (-23) + \dots\}; S_{36}$ Arith<br>$a_1 = -29, n = 36, d = 2$<br>$a_{36} = 2(35) - 29$<br>$= 41$<br>$S_{36} = 36 \left( \frac{-29 + 41}{2} \right)$<br>$= \boxed{216}$ |                                                                                     |
| <b>42.</b> $\sum_{y=1}^9 -2 \cdot (-3)^{y-1}$ Geo<br>$a_1 = -2, r = -3, n = 9$<br>$S_9 = \frac{-2(1 - (-3)^9)}{1 - (-3)} = \boxed{-9842}$                                                                          | <b>43.</b> $\sum_{n=1}^{24} (4n - 7)$ Arith<br>$a_1 = -3, a_{24} = 89, n = 24$<br>$S_{24} = 24 \left( \frac{-3 + 89}{2} \right)$<br>$= \boxed{1032}$                                                        |                                                                                     |
| <b>44.</b> $\sum_{k=3}^{10} 64 \cdot \left(\frac{1}{2}\right)^{k-1}$ Geo<br>$a_1 = 16, r = \frac{1}{2}, n = 8$<br>$S_8 = \frac{16(1 - (\frac{1}{2})^8)}{1 - \frac{1}{2}} = \boxed{\frac{255}{8}}$                  | <b>45.</b> $\sum_{m=6}^{39} (2m - 38)$ Arith<br>$a_1 = -26, a_{34} = 40, n = 34$<br>$S_{34} = 34 \left( \frac{-26 + 40}{2} \right)$<br>$= \boxed{238}$                                                      |                                                                                     |
| <b>Determine if the series is convergent or divergent. Find the sum, if possible.</b>                                                                                                                              |                                                                                                                                                                                                             |                                                                                     |
| <b>46.</b> $\left\{-\frac{3}{2} + \frac{3}{4} - \frac{3}{8} + \frac{3}{16} + \dots\right\}$ $r = -\frac{1}{2} \rightarrow \text{Conv.}$<br>$S_n = \frac{-\frac{3}{2}}{1 - (-\frac{1}{2})} = \boxed{-1}$            | <b>47.</b> $\left\{24 + 6 + \frac{3}{2} + \frac{3}{8} + \dots\right\}$ $r = \frac{1}{4} \rightarrow \text{Conv.}$<br>$S_n = \frac{24}{1 - \frac{1}{4}} = \boxed{32}$                                        |                                                                                     |

|                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>48. <math>\{4 + 16 + 64 + 256 + \dots\}</math></p> <p><math>r = 4 \rightarrow</math> <span style="border: 1px solid black; padding: 2px;">divergent</span></p>                                                                                                                                                                           | <p>49. <math>\{500 - 300 + 180 - 108 + \dots\}</math></p> <p><math>r = -3/5 \rightarrow</math> conv.</p> <p><math>S_n = \frac{500}{1 - (-3/5)} =</math> <span style="border: 1px solid black; padding: 2px;">312.5</span></p>                                                                                                                                                 |
| <p>50. <math>\sum_{k=1}^{\infty} 9 \cdot \left(\frac{4}{5}\right)^{k-1}</math> <math>r = \frac{4}{5} \rightarrow</math> conv.</p> <p><math>S_n = \frac{9}{1 - \frac{4}{5}} =</math> <span style="border: 1px solid black; padding: 2px;">45</span></p>                                                                                      | <p>51. <math>\sum_{x=1}^{\infty} 10 \cdot \left(-\frac{4}{3}\right)^{x-1}</math></p> <p><math>r = -\frac{4}{3} \rightarrow</math> <span style="border: 1px solid black; padding: 2px;">divergent</span></p>                                                                                                                                                                   |
| <p>52. <math>\sum_{n=4}^{\infty} -54 \cdot \left(\frac{1}{3}\right)^{n-1}</math> <math>r = \frac{1}{3} \rightarrow</math> conv.</p> <p><math>a_1 = -54 \left(\frac{1}{3}\right)^3</math></p> <p><math>= -2</math></p> <p><math>S_n = \frac{-2}{1 - \frac{1}{3}} =</math> <span style="border: 1px solid black; padding: 2px;">-3</span></p> | <p>53. <math>\sum_{i=5}^{\infty} -\frac{9}{2} \cdot \left(-\frac{1}{2}\right)^{i-1}</math> <math>r = -\frac{1}{2} \rightarrow</math> conv.</p> <p><math>a_1 = -\frac{9}{2} \left(-\frac{1}{2}\right)^4 = \frac{-9}{32}</math></p> <p><math>S_n = \frac{-9/32}{1 - (-1/2)} =</math> <span style="border: 1px solid black; padding: 2px;"><math>\frac{-3}{16}</math></span></p> |

#### Topic 6: Applications

| <p>54. Bricks are stacked in a pile with 42 bricks on the bottom row and 9 bricks on the top row. Each row has three fewer bricks than the row below it. Write a formula to represent the number of bricks on each row, then determine the number of bricks on the 7<sup>th</sup> row.</p> <p><math>a_1 = 42</math><br/> <math>a_n = 9</math><br/> <math>d = 3</math></p> <p><math>a_n = -3(n-1) + 42</math><br/> <math>= -3n + 45</math></p> <p><math>a_7 = -3(7) + 45</math><br/> <math>= 24</math></p> <p style="text-align: right;"><span style="border: 1px solid black; padding: 2px;">24 bricks</span></p>                                                                                                                                                            |       |       |   |    |   |    |   |    |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|---|----|---|----|---|----|--|
| <p>55. A company is offering a job with a salary of \$48,000 for the first year, then a raise of 2% each year after that. Write a formula to represent the salary after each year of employment, then determine the salary after 30 years.</p> <p><math>a_1 = 48000</math><br/> <math>r = 1.02</math></p> <p><math>a_n = 48000(1.02)^{n-1}</math></p> <p><math>a_{30} = 48000(1.02)^{29}</math><br/> <span style="border: 1px solid black; padding: 2px;"><math>= \\$85,240.55</math></span></p>                                                                                                                                                                                                                                                                             |       |       |   |    |   |    |   |    |  |
| <p>56. The table shows the number of seats in the first three rows of the concert hall. This pattern continues for a total of 12 rows. Determine if there are enough seats for a group of 500 people to attend a concert.</p> <table border="1" style="margin-bottom: 10px;"> <thead> <tr> <th>Row</th> <th>Seats</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>30</td> </tr> <tr> <td>2</td> <td>32</td> </tr> <tr> <td>3</td> <td>34</td> </tr> </tbody> </table> <p><math>a_1 = 30</math><br/> <math>n = 12</math><br/> <math>d = 2</math></p> <p><math>a_{12} = 2(12-1) + 30</math><br/> <math>= 52</math></p> <p><math>S_n = 12 \left( \frac{30 + 52}{2} \right)</math><br/> <math>= 492</math></p> <p style="text-align: right;">No, there are not enough seats.</p> | Row   | Seats | 1 | 30 | 2 | 32 | 3 | 34 |  |
| Row                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Seats |       |   |    |   |    |   |    |  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 30    |       |   |    |   |    |   |    |  |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 32    |       |   |    |   |    |   |    |  |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 34    |       |   |    |   |    |   |    |  |

Name: \_\_\_\_\_

Date: \_\_\_\_\_ Per: \_\_\_\_\_

# Unit 10 Test

## Sequences & Series

1. Give the first five terms of the sequence:

$$a_1 = -4; a_n = 2a_{n-1} - 1 \text{ (for } n \geq 2\text{)}$$

-4   -9   -19   -39   -79

2. Give the first five terms of the sequence:

$$a_1 = \frac{2}{5}, a_2 = \frac{4}{5}; a_n = a_{n-2} \cdot a_{n-1} \text{ (for } n \geq 3\text{)}$$

$\frac{2}{5}$     $\frac{4}{5}$     $\frac{8}{25}$     $\frac{32}{125}$     $\frac{256}{3125}$

3. Give the first five terms of the sequence:

$$a_n = n^3 - 2n$$

-1   4   21   56   115

4. Give the first five terms of the sequence:

$$a_n = 1 - \frac{1}{n}$$

0    $\frac{1}{2}$     $\frac{2}{3}$     $\frac{3}{4}$     $\frac{4}{5}$

5. Which formula can be used to represent the sequence below?

$$\{2, 5, 17, 65, 257, \dots\}$$

- A.  $a_n = n_3 - 3$   
 B.  $a_n = 4n - 3$   
 C.  $a_1 = 2; a_n = (a_{n-1})^3 - 3 \text{ (for } n \geq 2\text{)}$   
 D.  $a_1 = 2; a_n = 4a_{n-1} - 3 \text{ (for } n \geq 2\text{)}$

D

6. Which formula can be used to represent the sequence below?

$$\{-6, -3, 2, 9, 18, \dots\}$$

- A.  $a_n = 3n - 9$   
 B.  $a_n = n_2 - 7$   
 C.  $a_1 = -6; a_n = 2a_{n-1} + 9 \text{ (for } n \geq 2\text{)}$   
 D.  $a_1 = -6; a_n = \frac{1}{2}a_{n-1} + 1 \text{ (for } n \geq 2\text{)}$

B

7. Find the sum of the first nine terms in the series below.

$$\{1 + 8 + 27 + 64 + 125 + \dots\}$$

$$+ 216 + 343 + 512 + 729$$

2,025

8. Expand and evaluate the series:

$$\sum_{k=1}^{15} (5 - 2k^2)$$

$$3 - 3 - 13 - 27 - 45 - 67 - 93 - 123$$

$$- 157 - 195 - 237 - 283 - 333 - 387$$

$$- 445$$

-2,045

For questions 9-12, describe the sequence as arithmetic, geometric, or neither.  
For arithmetic and geometric sequences, give the common difference or common ratio.

9.  $\{3, -6, 12, -24, -48, \dots\}$

Type of Sequence

Common  
Difference/Ratio

Geometric

$r = -2$

10.  $\left\{-\frac{11}{4}, -2, -\frac{5}{4}, -\frac{1}{2}, \frac{1}{4}, \dots\right\}$

Type of Sequence

Common  
Difference/Ratio

Arithmetic

$d = \frac{3}{4}$

11.  $\{3600, 1440, 576, 230.4, \dots\}$

Type of Sequence

Common  
Difference/Ratio

Geometric

$r = \frac{2}{5}$

12.  $\{600, -300, 100, -25, \dots\}$

Type of Sequence

Common  
Difference/Ratio

Neither

—

13. Identify the formula for the sequence given below.

$\{11, 7, 3, -1, -5, \dots\} \quad d = -4$   
 $-4(n-1) + 11$   
 $-4n + 15$

A.  $a_n = 15 - 4n$

B.  $a_n = 7 - 4n$

C.  $a_n = 11 \cdot (-4)^n$

D.  $a_n = 11 \cdot (-4)^{n-1}$

A

14. Identify the formula for the sequence given below.

$\{76.8, 96, 120, 150, \dots\} \quad r = \frac{5}{4}$

A.  $a_n = 76.8 + \frac{4}{5}n$  C.  $a_n = 76.8 \cdot \left(\frac{4}{5}\right)^{n-1}$

B.  $a_n = 76.8 + \frac{5}{4}n$  D.  $a_n = 76.8 \cdot \left(\frac{5}{4}\right)^{n-1}$

D

15. Write a formula for the sequence below, then find  $a_{10}$ .

$\{-4, -12, -36, -108, \dots\} \quad r = 3$   
 $a_{10} = -4(3)^{10-1}$   
 $= 78732$

Formula

$a_n = -4 \cdot (3)^{n-1}$

$a_{10} = -78,732$

16. Write a formula for the sequence below, then find  $a_{45}$ .

$\left\{-2, -\frac{7}{4}, -\frac{3}{2}, -\frac{5}{4}, -1, \dots\right\} \quad d = \frac{1}{4}$   
 $a_n = \frac{1}{4}(n-1) - 2$   $a_{45} = \frac{1}{4}(45) - \frac{9}{4}$   
 $= \frac{1}{4}n - \frac{9}{4}$   $= 9$

Formula

$a_n = \frac{1}{4}n - \frac{9}{4}$

$a_{45} = 9$

17. If the information given below represents an arithmetic sequence, find  $a_1$ .

$$a_{14} = 16 \text{ and } d = -\frac{3}{2}$$

$$16 = -\frac{3}{2}(14-1) + a_1$$

$$16 = -19.5 + a_1$$

$$35.5 = a_1$$

$$a_1 = 35.5$$

18. If the information given below represents an arithmetic sequence, find  $a_{27}$ .

$$a_1 = 49 \text{ and } a_8 = 14$$

$$14 = d(8-1) + 49$$

$$14 = 7d + 49$$

$$-35 = 7d \rightarrow d = -5$$

$$a_{27} = -5(27-1) + 49$$

$$= -81$$

$$a_{27} = -81$$

19. If the information given below represents a geometric sequence, find  $a_1$ .

$$a_4 = -320 \text{ and } r = -4$$

$$-320 = a_1 \cdot (-4)^{4-1}$$

$$-320 = a_1 \cdot (-64)$$

$$5 = a_1$$

$$a_1 = 5$$

20. If the information given below represents a geometric sequence, find  $a_3$ .

$$a_1 = 3645 \text{ and } a_6 = 15$$

$$15 = 3645 \cdot r^5$$

$$a_3 = 3645 \cdot \left(\frac{1}{3}\right)^{3-1}$$

$$\frac{15}{3645} = r^5$$

$$a_3 = 405$$

$$\frac{1}{243} = r^5$$

$$\frac{1}{3} = r$$

$$a_3 = 405$$

21. Given the sequence below, find  $S_{19}$ .

$$\{30, 23, 16, 9, 2, \dots\}$$

$$a_{19} = -7(19-1) + 30$$

$$= -96$$

$$S_{19} = 19\left(\frac{30-96}{2}\right)$$

$$= -627$$

A. -573

B. -592

C. -627

D. -649

C

22. Given the sequence below, find  $S_8$ .

$$\{3, -12, 48, -192, \dots\}$$

$$S_8 = \frac{3(1-(-4)^8)}{1-(-4)}$$

$$= -39321$$

A. 65535

B. -65535

C. 39321

D. -39321

D

23. Given the sequence below, find  $S_7$ .

$$\{4374, 1458, 486, 162, \dots\}$$

$$S_n = \frac{4374(1-(\frac{1}{3})^7)}{1-\frac{1}{3}}$$

$$= 6558$$

A. 6558

B. 6736

C. 7014

D. 7252

A

24. Given the sequence below, find  $S_{25}$ .

$$\left\{1, \frac{5}{3}, \frac{7}{3}, 3, \frac{11}{3}, \dots\right\}$$

$$a_{25} = \frac{2}{3}(25-1) + 1$$

$$= 17$$

$$S_{25} = 25\left(\frac{1+17}{2}\right)$$

$$= 225$$

A. 208

B. 225

C. 240

D. 272

B

25. Find the sum:  $\sum_{n=1}^{34} (-17 + 8n)$

$$a_1 = -9$$

$$a_{34} = 255 \quad S_{34} = 34 \left( \frac{-9 + 255}{2} \right) = 4182$$

A. 3806

B. 4074

C. 4182

D. 4468

C

26. Find the sum:  $\sum_{x=2}^7 16 \cdot \left( \frac{5}{2} \right)^{x-1}$

$$a_1 = 40$$

$$n = 6$$

$$S_n = \frac{40(1 - (\frac{5}{2})^6)}{1 - \frac{5}{2}}$$

A. 6483.75

B. 6499.75

C. 16249.375

D. 2593.5

A

27. Which series do not have a sum? Check all that apply.

$$r = \frac{2}{7}$$

(divergent)

$$r = \frac{1}{4}$$

$$r = \frac{4}{3}$$

☐  $\{343 + 98 + 28 + 8 + \dots\}$

☐  $\{36 + 9 + \frac{9}{4} + \frac{9}{16} + \dots\}$

☒  $\sum_{k=1}^{\infty} -120 \cdot \left( \frac{4}{3} \right)^{k-1}$

☒  $\left\{ \frac{5}{3} - \frac{10}{3} + \frac{20}{3} - \frac{40}{3} + \dots \right\}$

☒  $\left\{ \frac{1}{2} - \frac{3}{4} + \frac{9}{8} - \frac{27}{16} + \dots \right\}$

☐  $\sum_{n=1}^{\infty} 180 \cdot \left( -\frac{1}{5} \right)^{n-1}$

$$r = -2$$

$$r = -\frac{3}{2}$$

$$r = -\frac{1}{5}$$

28. Find the sum:  $\left\{ 18 - 12 + 8 - \frac{16}{3} + \dots \right\}$   $r = -\frac{2}{3}$

$$S_n = \frac{18}{1 - (-\frac{2}{3})}$$

$$= 10.8$$

A. 10

B. 10.8

C. 48

D. 54

B

29. Find the sum:  $\sum_{m=1}^{\infty} -32 \cdot \left( \frac{1}{4} \right)^{m-1}$   $r = \frac{1}{4}$

$$S_n = \frac{-32}{1 - \frac{1}{4}} = \frac{-128}{3}$$

A.  $\frac{128}{5}$

C.  $\frac{128}{3}$

B.  $-\frac{128}{5}$

D.  $-\frac{128}{3}$

D

30. Mark got a new job with a starting salary of \$32,000. If he gets a yearly raise of 2%, find Mark's salary after 25 years.

$$a_1 = 32,000$$

$$n = 25$$

$$r = 1.02$$

$$a_n = 32000 (1.02)^{25-1} = 51469.99$$

$$\$51,469.99$$



31. An auditorium has 21 rows of seats. The first row has 18 seats, and each succeeding row has two more seats than the previous row. How many seats are in the auditorium?

$$a_1 = 18$$

$$n = 21$$

$$d = 2$$

$$a_{21} = 2(21-1) + 18 \\ = 58$$

$$S_{21} = 21 \left( \frac{18 + 58}{2} \right)$$

$$= 798$$

798 seats

32. Jalen is in a 24 hour dance-a-thon as part of a fundraiser. He raised \$50 in the first hour of the competition, then another \$60 in the second hour, then another \$72 in the third hour. If this pattern continues, what is the total amount of money he will raise if he stays for the entire competition?

$$a_1 = 50$$

$$r = 1.2$$

$$n = 24$$

$$S_n = \frac{50(1 - 1.2^{24})}{1 - 1.2}$$

$$= 19624.21$$

\$19,624.21

# CREDITS

I use clipart and  
fonts in my products by:



Art with Jenny K



Many thanks to these  
talented artists!